

Type of Terrestrial Data Requested

On Sunday, August 17, 2025 at 09:17:57 Mountain Daylight Time, you requested the following Terrestrial data from the Alberta Biodiversity Monitoring Institute (ABMI):

1. The Raw Trees & Snags dataset from all sample years.
2. The Methodology and Metadata describing above data set(s).

Spatial Distribution

The spatial distribution of the data you requested is:

1. Core site locations systematically distributed across the Alberta in a 0 site grid, of which 0 been surveyed to date using terrestrial sampling protocols.
2. Off-grid site locations, especially selected across Alberta for their unique or otherwise important environmental characteristics, of which 0 have been surveyed to date using terrestrial sampling protocols.

Delivered Files

1. T01A Site Physical Characteristics
(A_T01A_Site_Physical_Characteristics_AllYear_AB_17278477480514261984.csv-1940KB)
has 22952 rows
2. T01B Site Suitability (A_T01B_Site_Suitability_AllYear_AB_15450932280384439475.csv-14603KB) has 139338 rows
3. T01C Site Capability (A_T01C_Site_Capability_AllYear_AB_13227002129353905960.csv-3972KB) has 57069 rows
4. T01D Ground Cover (A_T01D_Ground_Cover_AllYear_AB_15061258059849914594.csv-1535KB) has 24520 rows
5. T01E Site Disturbance (A_T01E_Site_Disturbance_AllYear_AB_182254224868863848.csv-2486KB) has 43083 rows
6. T02A Surface Substrate
(A_T02A_Surface_Substrate_AllYear_AB_2135933395461702043.csv-2926KB) has 50214 rows
7. T02B Surface Substrate Disturbance
(A_T02B_Surface_Substrate_Disturbance_AllYear_AB_12654151113688032006.csv-165KB)
has 3169 rows
8. T09 Trees, Snags and Stumps
(A_T09_Trees__Snags_and_Stumps_AllYear_AB_4161269286345444842.csv-16253KB) has 143464 rows
9. T10A Tree Core Ages (2007-2015)
(A_T10A_Tree_Core_Ages__20072015__AllYear_AB_16731053215387479356.csv-1486KB)
has 13102 rows
10. T10B Tree Damages (A_T10B_Tree_Damages_AllYear_AB_13720245095738923949.csv-1427KB) has 16111 rows
11. T10C Tree Core Rings (since 2016)
(A_T10C_Tree_Core_Rings__since_2016__AllYear_AB_4679532420702741635.csv-357KB)

has 2997 rows

12. T11 Canopy Closure (A_T11_Canopy_Closure_AllYear_AB_4378542045886589865.csv-740KB) has 16904 rows

Attention

1. Ten sites in the data you requested were sampled in 4 consecutive years. Normally, the ABMI surveys sites once every 5 years. The inclusion of 4 consecutive years of data from 10 sites may cause confusion during data analysis.
2. Sites surveyed during 2003-2006 (inclusive) were selected by ABMI to ensure that the protocols worked appropriately across a gradient of ecosystem types and human disturbances. Thus, the data you requested may not be entirely representative of landscapes from which the data was collected (i.e., not a random sample of ABMI's data points).

Terms of Use

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Raw breeding bird data (2004-2006 inclusive) from the Alberta Biodiversity Monitoring Institute was used, in whole or part, to create this product. More information on the Institute can be found at: <http://www.abmi.ca>.

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1. the version and type of ABMI data used,
2. if appropriate, the type and source of additional, non-ABMI data used in analysis,
3. a full description of data analysis, and
4. the person(s) or organization(s) conducting the analysis.

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Contact Information

If you have questions or concerns about your data request, please contact:

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Terrestrial Survey Methods

We had our sample design and methods extensively peer reviewed by the greater scientific community to ensure scientific credibility. Starting in May, through to the end of June we implemented spring terrestrial data collection protocols at the annually selected subset of the 1656 ABMI permanent monitoring sites, to sample breeding birds, soil arthropods (springtails and mites), site capability, physical characteristics, moss, lichen, downed woody materials and trees. Starting at the end of June through to the first week in August, we implemented summer terrestrial data collection protocols at each site to assess vascular plants, tree ages, and surface substrate (detailed terrestrial data collection protocols are available from the [ABMI website: ABMI 10001—Terrestrial Data Collection Protocols](#)). We implemented terrestrial protocols in the same way at all sites in each sampling year, except where protocol updates are noted.

Terrestrial Survey Sites

We characterized habitat suitability, ecosite capability, ground cover, surface substrate, and site disturbance at site centre, or bird point count station #1, and at additional stations (Figure OB00004) surrounding site centre.

Protocol Update: In 2015, remote camera traps and autonomous recording units (ARUs) were deployed at bird point count stations 3, 5, 7 and 9. We characterized physical characteristics, site suitability, capability and disturbance within a 50 m radius rather than 150 m radius for these sites. Collection methodology indicates the method used to collect ecosite data.

Physical Characteristics

[1 file](#)

We determined elevation using hand-held GPS units, and site slope and aspect using a hand-held compass. We identified the ABMI site centre in the field using hand-held GPS units. We used site centre as the reference point for all subsequent terrestrial protocols. We use the term “central hectare” to describe the 100×100 m area centered on site centre. We use the term “quadrant” to describe the four 0.25 ha (50×50 m) plots that subdivide the central hectare and originate from site centre.

Protocol Update: Data from 2003–2007 include elevation, slope and aspect at site centre (point count station 1) only. Data from 2008 do not include elevation, and slope and aspect were collected at each of the 9 point count stations.

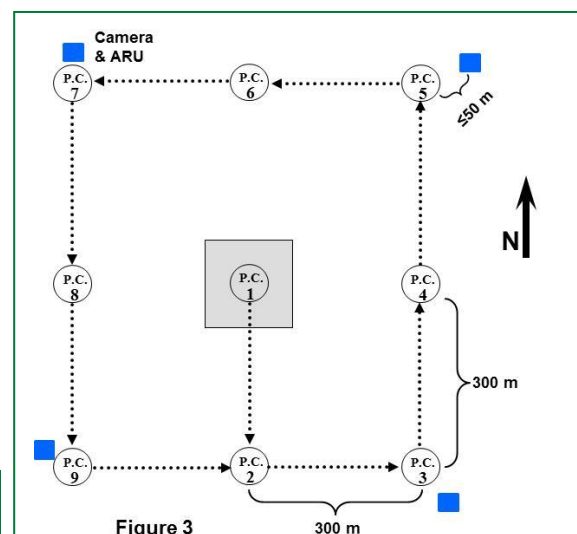


Figure OB00004. Point count, and Camera & ARU stations for mammal and breeding bird surveys at terrestrial monitoring sites.

Site Capability

[1 file](#)

Site capability was used to broadly describe the ecological habitat potential at each site and was described using dominant vegetative communities (ecological site classification; Beckingham and Archibald 1996). We modified Beckingham and Archibald (1996) to derive generalized eco-type categories (Table OB00002). We classified the primary and secondary ecosite/structural stage within a 150 m radius around each bird point count station and estimated the proportion of each ecosite in 10% increments. We only recorded secondary ecosites if it occupies >20% of the area. Secondary ecosites must make up ≥ 0.1 ha of a continuous habitat (i.e., minimum 35 m in diameter). The sum of primary and secondary ecosite proportions could not add to more than 100%.

Protocol Update: Data from the Prototype Rotation (2003–2006) include a primary and secondary ecological site (ecosite) classification for current conditions. Data from 2007–2011 includes the primary ecological site classification for current conditions, and primary and secondary ecosites for historical conditions. Data from 2012 and onwards includes historical and current conditions for primary and secondary ecosites.

In 2015, we deployed autonomous recording units (ARUs) at some sites and remote camera traps at all sites. Where ARUs and remote camera traps were deployed, primary and secondary ecosite/structural stage was classified within a 50 m radius around each ARU and remote camera trap location. Given the difference between bird point count and ARU collected ecosite information, we indicate the collection method used to collect ecosite data. These methods include bird point count points, ARU points, linear off-grid points, and off-grid ARU points. The latter three methods classify ecosite within 50 m of the ARU and remote camera trap.

To classify the ecosite, we first determined whether the location was upland or wetland. We then determined the moisture and nutrient category based on the understory vegetative community. We assigned a tree species modifier and structural stage to the classification; in non-tree-dominated structural stages, we used the “none” modifier. When the location was altered by human disturbance (e.g., agriculture, well pad, road, cutline, etc.), we based the classification on data collected outside the quadrant; therefore, the ecosite classification was based on pre-disturbance conditions (i.e., original stand conditions).

Table OB00002. ABMI Simplified Ecological Site Classification (Site Capability)

Dominant Shrub/Herb/Ground Cover	Nutr. / Moist. Code ¹	Tree Species Modifier	Tree Species Composition ² (In an area without human disturbance)	Structural Stage ³
Upland Vegetation Communities				A. Tree Dominated Ecosites (Trees ≥10% cover) – Add 4-letter code combining tree height, density, and arrangement. <u>Tree Height</u> (TS) Short – ≥50% of canopy cover <10 m tall. (TT) Tall – >50% of canopy cover ≥10 m tall. <u>Tree Density</u> (D) Dense – Trees ≥1.3 m tall are ≤2 m apart. (S) Sparse – Trees ≥1.3 m tall are >2 m apart. <u>Tree Arrangement</u> (C) Complex (Spatially) – Tallest trees ≥10 m apart, with smaller trees (~ ½ height) between that receive direct sunlight from above. (N) Non-complex (Spatially) – Tallest trees <10 m apart, with few or no smaller trees (~ ½ height) between, that receive direct light from above. B. Non-Tree Dominated Ecosites (Trees <10% cover) Shrubs Present (Shrubs >10%; Trees <10%) – Add 3 letter code combining shrub height and density. <u>Shrub Height</u> (SL) Low – Shrubby vegetation <2 m tall (ST) Tall – Shrubby vegetation >2 m tall <u>Shrub Density</u> (D) Dense – Shrubs cover >75% (M) Moderate – Shrubs cover 25-75% (S) Sparse – Shrubs cover <25% Small Trees Present <u>Small Tree Height</u> (TL) Low – Small trees <1.3 m tall <u>Small Tree Density</u> (D) Dense – Small trees cover >75% (M) Moderate – Small trees 25-75% (S) Sparse – Small trees <25% Only Ground Vegetation Present (Shrubs <10%; Trees <10%; Other Vasc. >10%) – Add 3-letter code combining dominant vegetation type and density <u>Vegetation Type⁷</u> (GB) Bryoid/Lichen – Bryophyte and lichen (GF) Forb – Non-graminoid herbs and ferns (GG) Graminoid – grasses, sedges (GR) Marsh – reeds, rushes, and cattails <u>Vegetation Density</u> (D) Dense – Cover >75% (M) Moderate – Cover 25-75% (S) Sparse – Cover <25% Non-Vegetated (<10% Vegetation Cover) – Add 2-letter code describing dominant substrate type. (NR) – Bedrock, cliff, talus, bolder (NS) – Sand bar in river/stream (cobble, gravel, sand) (NB) – Beach at edge of a lake or wetland (NM) – Mineral soil any other reason (NO) – Organic soil any other reason Note: If standing water is present, refer to Open Water Communities
Poor – Xeric Bearberry, Lichen, Bog Cranberry common at some sites	1 - PX	1a Pine	Pj + Fd > 80%	
Poor – Mesic Labrador Tea, Feather Moss, Bog Cranberry, Bilberry, Grouse-berry common at some sites	2 - PM	2a Pine	Pj + Pl > 50%	
		2b Other	Aw + Sw + Se + Fa +Pw > 50%	
		2c Sb	Sb > 50%	
Medium – Xeric Hairy Wild Rye, Bearberry, Canada Buffalo-berry, Feather Moss common at some sites	3 - MX	3a None	No Trees	
		3b Pine	Pj + Pl > 50%	
		3c AwMix	Aw > 20%	
		3d Spruce	Sw + Se + Fa >50%	
Medium – Mesic Low-bush Cranberry, Canada Buffalo-berry, Blueberry, Rose, Alder, Labrador Tea, Bearberry, Thimbleberry, Bog Cranberry, Feather Moss common at some sites	4 - MM	4a Pine	Pj + Pl + Fa >50%	
		4b PjMix	Aw + Bp + Sw >20%, AND Pj >20%	
		4c Aw	Aw > 50%	
		4d AwMix	Aw >20% AND Sw + Sb + Pl > 20%	
		4e Spruce	Sw > 50%	
Medium – Hygric Horsetail, Dogwood, Rose, Willow, Feather Moss common at some sites	5 - MG	5a Poplar	Pb + Aw > 50%	
		5b Spruce	Sw + Se > 50%	
		5c Sb	Sb > 50%	
Rich – Hygric Dogwood, Fern, Feather Moss, Rose, Alder, Bracted Honeysuckle, Devil’s Club Fir common at some sites	6 - RG	6a Pine	Pl > 50%	
		6b Poplar	Pb + Aw > 50%	
		6c Spruce	Sw + Se + Fa > 50%	
Not Treed	7 - NT	7a Alpine	Elevation above tree line	
		7b Flood ⁴	Little to no vegetation present due to frequent flooding	
		7c Ice	Site disturbed frequently by ice or snow	
		7d Dry	Site in prairies/parkland and receives little precipitation	
		7e Geo	Geological features not suitable for tree growth	
		7f Human ⁵	Site disturbed recently by humans	

Dominant Shrub/Herb/Ground Cover	Nutr./Moist. Code ¹	Tree Species Modifier	Tree Species Composition ² (In an area without human disturbance)	Structural Stage ³
Lowland/Wetland Vegetation Communities				C. Open Water Dominated Communities (Emergent Vegetation <10%) – Add 4-letter code combining dominant vegetation type, height and density <u>Vegetation Type</u> (OV) Vegetated – Floating or submerged plants ≥ 10% cover (ON) Non-Vegetated – Floating or submerged plants < 10% cover (note that only a 2-letter code is used for this category) <u>Vegetation Height</u> (S) Short Submerged – ≥50% of vegetation extending 0.0 – <0.3 m above the substrate (M) Medium Submerged – ≥50% of vegetation extending 0.3 – 1.3 m above the substrate (T) Tall Submerged – ≥50% of vegetation extending >1.3 m above the substrate (F) Floating – ≥50% of vegetation with floating leaves on the water surface. <u>Vegetation Density</u> (D) Dense – Aquatic vegetation covering >75% of the substrate. (M) Moderate – Aquatic vegetation covering 25-75% of the substrate. (S) Sparse – Aquatic vegetation covering <25% of the substrate. <u>Tree Species Codes:</u> Aw – trembling aspen Pb – balsam poplar Pc – plains cottonwood Bp – paper birch Ba – Alaska birch Pl – lodgepole pine Pj – jack pine Pw – white pine Sb – black spruce Se – Engelmann spruce Fa – subalpine fir Fd – Douglas fir Fb – balsam fir Lt – larch Mm – Manitoba maple Am – western mountain ash Sw – white spruce
Bog: Poor – Hydric Labrador Tea, Peat Moss, Lichen, Bog cranberry and Cloudberry may also be present (Soil saturated for part or all the year. Undecomposed organic soil substrate)	8 - PD	8a Sb ⁶	≥10% tree cover (may only be in shrub/ground strata) Sb > 50%	
		8b Shrub	<10% tree cover	
Poor Fen: Medium – Hydric Labrador Tea, Peat Moss, Sedge, Bog cranberry, Dwarf Birch and Willow may also be present (Soil saturated for part or all the year. Undecomposed organic soil substrate)	9 - MD	9a SbLt	≥10% tree cover (may only be in shrub/ground strata) Sb + Lt > 50%	
		9b Shrub	<10% tree cover	
Rich Fen: Rich – Hydric Dwarf Birch, Willow, Sedge, Grass, Moss, (Soil saturated for part or all the year; undecomposed organic soil substrate; includes floating mats of vegetation)	10 - RDp	10a SbLt	≥10% tree cover (may only be in shrub/ground strata)	
		10b Shrub	<10% tree cover AND ≥10% shrub cover	
		10c None	<10% tree cover AND <10% shrub cover	
Wet Meadow: Rich – Hydric Dominated by sedge, grass, presence of shrub and trees (e.g. willow). Conductivity < 15 (Soil saturated for part or all of the year. Well decomposed, organic soil substrate.)	10.5 - RDM	10.5a Tree	≥10% tree cover (usually along wetland edge, may only be in shrub/ground strata)	
		10.5b Shrub	<10% tree cover AND ≥10% shrub cover	
		10.5c None	<10% tree cover AND <10% shrub cover	
Marsh: Very Rich – Hydric Cattail, Rush, Reed, Conductivity < 15 mS/cm, sedge and grass may also be present (Water is above the rooting zone for most or all of the year)	11 - VD	11a None	Usually along a water body edge ≥10% emergent vegetation cover <10% tree cover	
Swamp Trees and shrubs present, poorly developed bryophytes, often with pools of water (Water is above the rooting zone for some of the year, mineral or humified organic soil rather than peaty)	12 - SD	12a Tree	>10% tree cover	
		12b Shrub	<10% tree cover	
Alkali Conductivity > 15 mS/cm, white salt flats at water's edge (Water is above the rooting zone for most or all of the year)	13 - AD	13a None	<10% shrub/tree cover	
Open Water	14 - OW	14a Lake	In standing water <10% emergent cover	
		14b River	In flowing water <10% emergent vegetation cover	

Classifications are based on Dominant Shrub/Herb/Ground Cover before determining the Tree Species Modifier and Structural Stage. Tree species compositions in the tables are the “simplified categories” for the ABMI - these may not fit perfectly with what is seen at the site.

1. Note that moisture nutrient category names are approximate and the category often also includes adjacent nutrient and moisture categories (Nutrient Status: P=Poor, M=Medium, R=Rich, V=Very Rich; Moisture Status: X=Xeric, M=Mesic, G=Hygic, D=Hydric, OW=Open Water. NT, SD, AD are exceptions).
2. Tree species composition is determined from both the dominant/co-dominant (canopy) and intermediate/suppressed (sub-canopy) trees, giving more weight to the dominant and co-dominant trees.
3. Determine the structural stage by first determining if the site is tree-dominated, non-tree dominated, or open-water dominated after ecological-site type is determined. Then choose the appropriate code combination paying careful attention to the descriptors.
4. Use 7b (NT-Flood) for sites at the edge of rivers, streams, lakes and wetlands where vegetation is disturbed frequently by flooding. The area is either non-vegetated or dominated by grasses, sedges and forbs, with trees/shrubs absent. Note that areas with water present seasonally, often with small permanent pools, but with trees/shrubs present, are classified as Swamp.
5. Use category 7f (NT-Human) only when other ecosite classifications are not appropriate. Note that NT-Human CANNOT be used for historic conditions.
6. Poor Fens are often black spruce (Sb) dominated and do not always contain Larch/Tamarack (Lt). The absence of Larch does not indicate that the site is PD – it could still be MD. Differentiation between PD and MD must be determined based on the understory species (i.e., presence of cloudberry and lichen in PD, with the addition of sedge, dwarf/bog birch and willow for MD).
7. Use GR for tall, typically emergent, vegetation including cattails, hardstem and softstem bulrush, giant reed grass (*Phragmites australis*), bur-reeds and small fruited bulrush.

Site Suitability

[1 file](#)

Site suitability was used to broadly describe the current habitat conditions in a 150 m radius around each bird point count station [2003-2015] or in a 50 m radius around locations where ARUs and/or remote camera traps were deployed as well as any bird point count stations [2015 onwards]. We recorded characteristics of veteran trees, dominant and co-dominant (canopy trees), and intermediate/suppressed (sub-canopy) trees, dominant shrub species and total shrub cover. We estimated total percent cover of shrubs in four categories (<25%, 25-50%, 51-75%, >75%). We estimated tree species composition (in 10% increments), average distance between trees (in 1 m increments), and average height of trees (to the nearest 5 m category) for all trees (excluding dead trees) within the following stratification categories:

- **Veteran Trees**—Trees that are considerably older than rest of the stand, usually remnant from a previous forest.
- **Dominant and Co-dominant Trees (Canopy Trees)**—Trees with well-developed crowns extending slightly above the general level of surrounding trees (dominant trees), receiving full light from above and partly from the side and/or trees with crowns (slightly smaller than dominant and crowded from the sides) forming the general level of surrounding trees, receiving full light from above and little from the side.
- **Intermediate and Suppressed Trees (Sub-canopy Trees)**—Trees with crowns (usually small and quite crowded) below, but extending to, the general level of surrounding trees, receiving little light from above and none from the sides and/or trees with crowns entirely below the general level of the surrounding trees, receiving no direct light from above or the side.

Protocol Update: Data from the Prototype Rotation (2003–2006) includes an estimated rank abundance for the top 4 most abundant species for each canopy layer. This information was not collected after 2006.

Site Disturbance

[1 file](#)

At each site, we recorded human-caused disturbances within a 150 m radius around bird point counts stations or within a 50 m radius of locations where ARUs and/or remote camera traps were deployed. Human-caused disturbances included: harvest (e.g., clear-cut, partial-cut, understory retention, etc.), linear disturbance (e.g., cutline, seismic, pipeline, powerline, etc.), energy well pad, roads or trails, agricultural disturbance, urban or residential development, industrial development, and bare ground for which the cause cannot be determined. We also recorded natural disturbances including evidence of fire, wind damage, ice, beaver, erosion, flooding, insect damage, and diseases.

Protocol Update: Data from the Prototype Rotation (2003–2006) include Rank of Disturbance where more than one disturbance type was noted. Data collected in 2007 and later does not include rank of disturbance. See [ABMI Terrestrial Data Collection Protocols](#) for detailed information.

For data from 2013-2015 human disturbance was not recorded, only natural disturbance was recorded.

Ground Cover

[I file](#)

Within a 150 m radius around each bird point count station or within a 50 m radius of each ARU/remote camera trap, we recorded characteristics of herbaceous cover and type of ground cover. The method used is indicated by collection methodology. We estimated percent cover for dead herbaceous cover in four categories (< 25%, 25%–50%, 51%–75%, and > 75%). For dead herbaceous vegetation, we also determined the predominant height. Finally, we identified the most common ground vegetation by functional group (grass, herbs, moss, lichens, and shrubs), and estimated the percent cover of bare ground and water in 5% increments.

Protocol Update: Percent cover of bare ground and water data recorded as 0, <1% and then in 5% increments from 2013 onward.

Surface Substrate

[I file](#)

We measured the depth of organic content using two 30 m transects paralleling the NE and SW quadrant boundaries (Figure OB00003). We started these transects 55 m north and south of the site centre, and sampled in a clockwise direction to ensure soil inside the quadrant was not disturbed. We measured the depth of organic matter and/or buried wood every 2 m along each 30 m transect (totaling 15 sampling points per transect). Using a planter's shovel and measuring stick, we recorded the depth of organic matter, to the nearest 0.5 cm, to mineral soil, or other impenetrable substrate, such as rock or a frozen layer.

We defined organic matter as the litter, fermented, and humus (LFH) layer of the soil horizon. We defined buried wood as downed wood in all decay stages (see Downed Woody Material in [ABMI 10001—Terrestrial Data Collection Protocols](#)) that is >50% below the ground surface. If organic matter had developed over the wood (i.e., decay stage 5) and the wood was <10 cm thick, we classified it as organic matter. In areas where the LFH was indistinct, such as grasslands, we included the plant roots in the LFH. When no LFH layer was visible (e.g., cultivated agriculture areas), we only measured the litter layer (if present). In lowland areas (e.g., black spruce bogs) where organic matter was deeper than 40 cm, a 5 m long soil probe was used to determine the depth of LFH. We recorded depth every 4 m along each transect and recorded depths in 10 cm increment after 40 cm. When sample points were positioned directly on rock or over standing water with a surface area greater than 1 m², we took no measurements. When rocks or water were less than 1 m², or the sample point landed on an impenetrable downed log or stump (above the ground surface) we took measurements directly perpendicular and to the right-hand

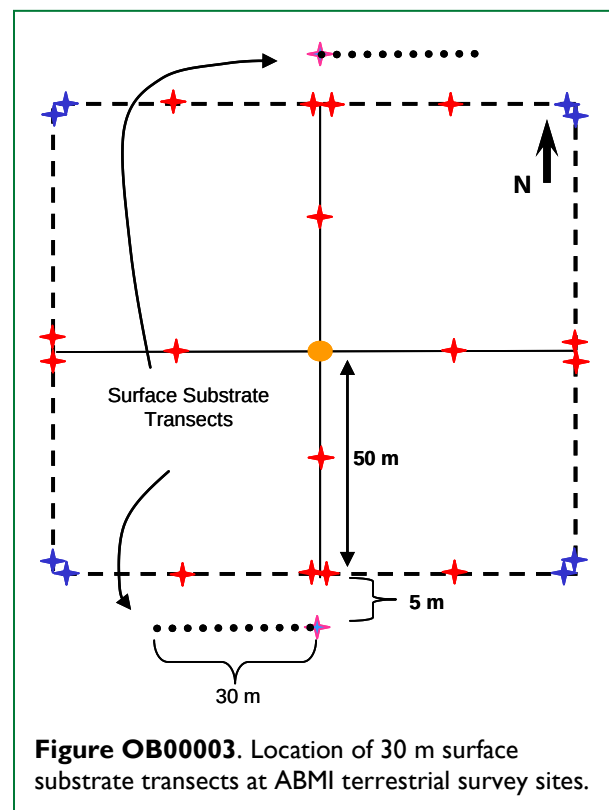


Figure OB00003. Location of 30 m surface substrate transects at ABMI terrestrial survey sites.

side of the intended sample point. Finally, we indicated when only partial measurements could be obtained (e.g., hit frost, root, or rock).

At the start of each transect, we described the slope position:

- Crest—situated in a relatively level area on the top of a hill;
- Slope—situated on the side of a hill and including a steepness modifier (1 for slopes 2°–5°; 2 for slopes 6°–10°; 3 for slopes 11°–30°; and 4 for slopes >30°);
- Toe—situated at the bottom of a hill where the ground surface transitions from a slope to level;
- Level—situated in an area with <2° slope, and;
- Depression—situated in an area that accumulates water after rain events.

We recorded slope aspect (looking down hill) for all plots.

We also measured human and natural disturbance in the 30 m x 2 m plot [2009 onwards]. We noted cause of disturbance using the disturbance categories from the site suitability protocol and the percent area affected.

Sampled Sites and Collection Methodology

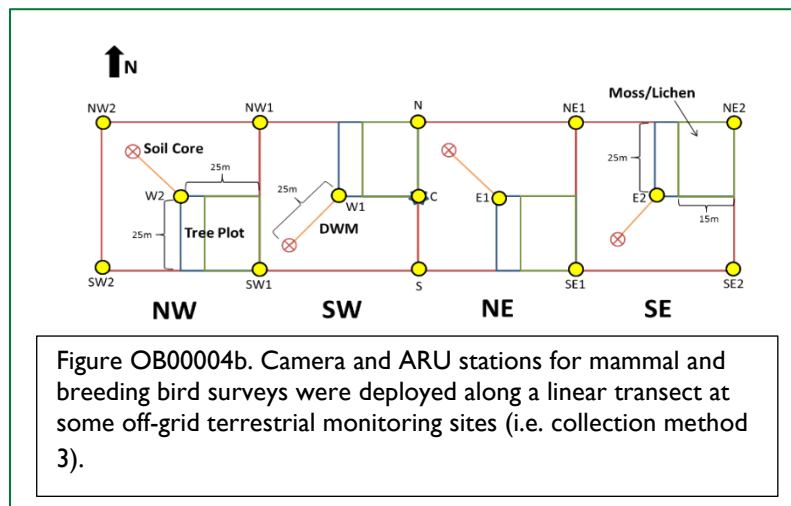
The ABMI has been surveying systematic grid sites since 2003.

In 2015 and 2017 we surveyed off-grid sites along linear features. At these locations, sampling plots were oriented in a linear fashion along the targeted linear features.

Given the difference between bird point count and ARU collected ecosite information, we indicate the collection method used to collect ecosite data (Figure OB00004). These methods include:

1. Collection method 1: bird point count station points.
2. Collection method 2: ARU points.
3. Collection method 3: Linear off-grid ARU points (Figure OB0004b).
4. Collection method 4: Targeted off-grid ARU points.

The latter three methods classify ecosite within 50 m of the ARU and remote camera trap.



Trees, Snags and Stumps

I file

We counted trees, snags, and stumps at each ABMI site using four 25 × 25 m plots with two nested sub-plots (10 × 10 m and 5 × 5 m; Figure OB00017). We anchored the four subplots at 35.35 m from site centre in each of the four sub-ordinal directions (i.e., NE, SE, SW, NW). In the 5 × 5 m sub-plots, we recorded all trees and snags ≥1.3 m in height as well as stumps. In the 10 × 10 m sub-plots, we recorded all trees and snags ≥7 cm diameter at breast height (dbh) as well as stumps. In the 25 × 25 m plots, we recorded all trees and snags ≥25 cm dbh as well as stumps.

Protocol Update: Data from 2003-2007, only trees ≤7 cm dbh were recorded in the 5 × 5 m sub plot. Height (m) was also recorded in all plots for all trees, snags, and stumps. Data from 2003-2005 did not report a crown class, top height or base height.

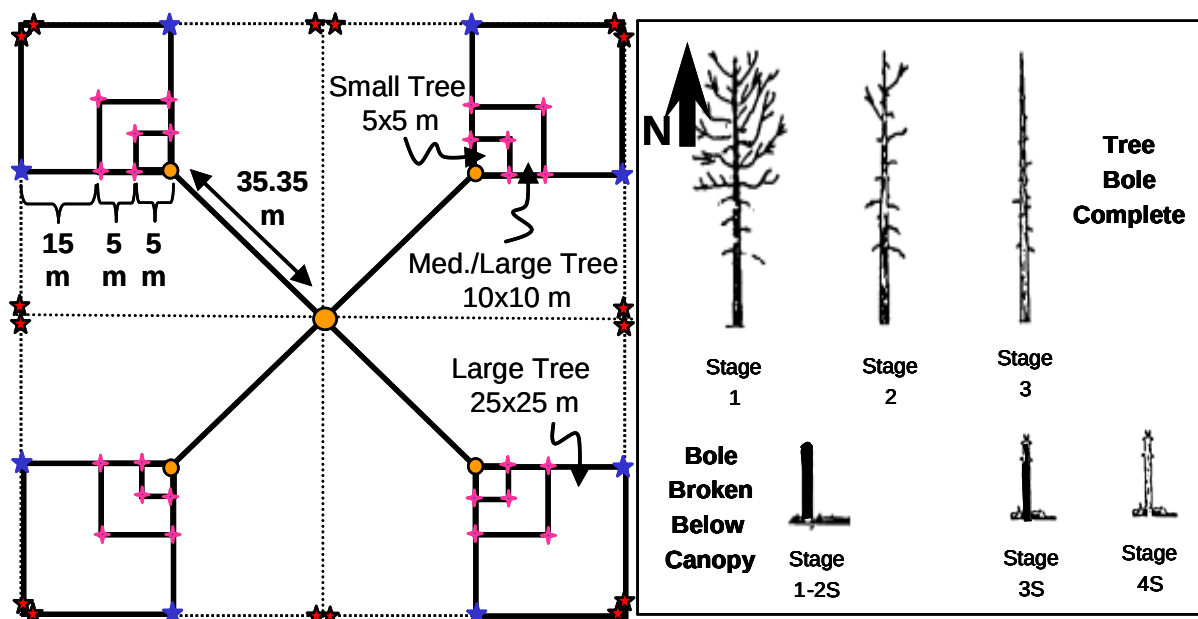


Figure OB00017. Plot layout for tree, snag, and stump measurements at ABMI terrestrial monitoring sites and decay classes for snags.

We recorded species, dbh (cm), condition, and decay stage of all trees, snags, and stumps. For all trees with a dbh ≥7 cm we also recorded height to crown and base, and crown class. We measured the height of the crown top and crown base for 9 live trees when more than 10 live trees were present in the 10 × 10 m sub-plot; we estimated heights for the remaining trees. We further classified trees with a dbh ≥7 cm into five crown classification categories:

- 1) Veteran trees, which are older than the rest of the stand and usually a remnant from a previous forest;
- 2) Dominant trees, which have well-developed crowns extending slightly above the canopy of the surrounding trees;
- 3) Co-dominant trees, which are slightly smaller than dominant trees and at the general height of surrounding trees;
- 4) Intermediate trees, which have crowns below the average canopy height but extending to the general level of the surrounding trees, and;

5) Suppressed trees, which have crowns entirely below the general level of the surrounding trees.

We defined snags as ≥ 1.3 m tall and leaning no more than 45° from perpendicular, and stumps as < 1.3 m in height. There is no minimum diameter for stumps. We classified snags into one of three decay stages:

- 1 = Recently dead, all twigs/branches present, wood hard, bark (normally) intact;
- 2 = Twigs and small branches missing (major branches remain), wood hard, and;
- 3 = No branches, bole mostly intact, wood starting to soften (Figure OB00017).

If the snag was broken below the canopy and no branches and twigs were present to evaluate decay class, we evaluated decay class based on the condition of the remaining stump:

- 1-2S—recently dead, wood hard, bark (normally) intact;
- 3S—wood starting to soften;
- 4S—wood soft throughout the snag.

We recorded tree species (if possible), height, and inside diameter of each stump.

Protocol Update: Between 2003 and 2006, we measured trees and snags with a dbh ≥ 25 cm in a 25×50 m plot. 2007 and onward, we measured these large trees in a 25×25 m plot. The area in the 25×25 m plots used in 2007 and onwards do not overlap with the plots used between 2003 and 2006.

Tree Ages

[1 file](#)

We obtained tree cores from a maximum of 9 trees in the central hectare to estimate age and growth rate of trees at each AMBI monitoring site. We obtained a tree core using a 16-inch increment borer and recorded the height (using a vertex hypsometer) of the largest (determined using diameter at breast height (dbh)) “live” tree within the central hectare, regardless of species. We obtained tree cores and tree heights from the largest “live” trees from the leading species (species with the highest density) of dominant and/or co-dominant (canopy) trees within quadrants (NE, SE, SW, NW) of the central hectare. We did not sample veteran or residual trees from a former stand. We also obtained tree cores and tree heights from the largest “live” trees from the secondary species (species with the 2nd highest density of stems) of dominant and/or co-dominant (canopy) trees within quadrants. We sampled the secondary tree species only when they comprised >20% of stem density within a specific quadrant.

For trees cored, we recorded any significant tree damage that could affect tree height, such as: broken tops, dead tops, forks, crooks, and/or abnormal scarring or other damage (e.g., mistletoe). We collected tree cores facing toward site-centre when possible, boring the trees at 1.3 m from the base. When trees were <10 cm dbh, we destructively sampled a representative tree from outside the quadrant by taking a cookie. When all trees were <10 cm dbh we only obtained a cookie from the leading tree species from outside of the central hectare, sampling a total of 4 trees per site. We preserved tree cores in straws and stapled straw ends shut to allow the core to dry and prevent the core from decaying and mold from forming. Holes were cut along the length of the straws to allow for air circulation. We stored samples in a dry area before final processing in the laboratory where we counted rings of the cores and archived the samples.

From 2007 to 2015, cores were analyzed by the Canadian Forest Service for the National Forest Inventory (NFI) in a laboratory setting. Tree cores are mounted (glued) onto prefabricated mounting boards allowing them to dry for 2 days before sanding. Using a computer and specialized software, the number of tree rings present (excluding the bark) are counted. Cookies from small trees were not mounted; however, we sanded and counted the rings similar to core samples. The finalized data is then stored at the Canadian Forest Service. Additionally, the tree core data is amalgamated with tree age data and dbh of similar-height trees from tree data collected at our sites.

From 2016 onward, cores were analyzed by the Monitoring Centre in a laboratory setting. Tree cores and cookies are oven dried at 30°C for 48 hours prior to mounting. All tree cores are then mounted (glued) onto prefabricated mounting boards allowing the glue to dry before sanding. Using a computer, GIMP Image Editing Program, and CooRecorder a simplified count of the number of tree rings present (excluding the bark) is completed. Cookies from small trees are not mounted; however, we sanded and counted the rings similar to core samples. We also note the quality of the tree core collected when processing. The finalized data is amalgamated with tree age data and dbh of similar-height trees from tree data collected at our sites.

Canopy Closure

I file

We measured canopy closure at 35.35 m and 49.49 m from the site centre in each of the four sub-ordinal directions (i.e., NE, SE, SW, NW) for a total of 8 readings per site (Figure OB00018). Using a spherical (concave) densiometer placed 1.3 m above the ground and leveled to ensure constant heights for each measurement, we recorded canopy cover readings at 35.35 m facing towards site centre, and readings at 49.49 m facing away from site centre. Canopy closure values range from 0 (high canopy closure) to 96 (no canopy closure).

Protocol Update: Between 2003 and 2005, we measured canopy closure at 30 and 60 m from site-centre in each of the sub-ordinal directions, rather than at 35.35 and 49.49 m as described above.

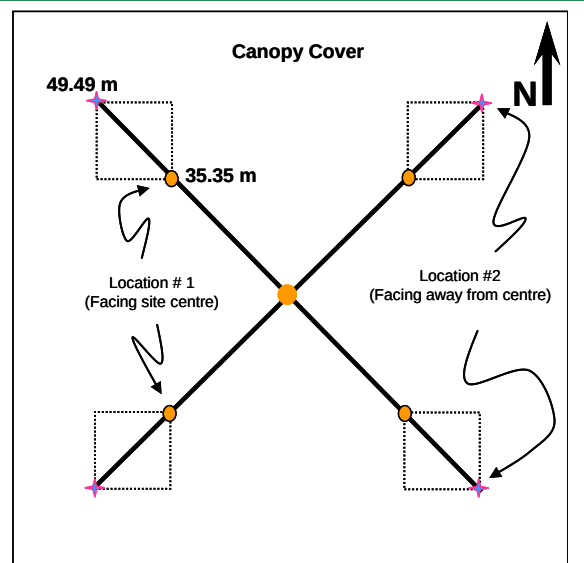


Figure OB00018. Locations of canopy cover readings at ABMI terrestrial monitoring sites.

Metadata

Metadata is simply “information about data” (Michener 2006). While there may be different levels of detail and structure to metadata, at its most basic, metadata provides the information required to understand a particular data set (e.g. structure and content) to ensure its accessibility (Michener et al. 1997; Michener 2006). This appendix provides the metadata for the data files that you requested from the Alberta Biodiversity Monitoring Institute.

General Codes

Throughout ABMI raw data files, the following codes and definitions are applied.

None or 0

None or “0” is applied to any variable that **was examined** by field crews and found to be absent. “None” is used for text entries and “0” is used for numerical entries. For example, when field crews examine the canopy and find no “Veteran” trees in the canopy, this is recorded as “None.” When there is no slope at the survey site, slope is recorded as “0.” The numeral “0” can also be used as a nominal code—for example, wind conditions can be recorded as “0”.

VNA—Variable Not Applicable

Some ABMI data is collected in a nested manner. For example Tree Species is a parent variable. This variable has a number of child variables that are used to describe the parent variable in more detail (e.g., condition, DBH, decay stage). When the parent variable is recorded as “None,” child variables are no longer applied and are recorded as “VNA.” VNA is also used when the protocol calls for a modified sampling procedure based on site conditions (e.g., surface substrate protocol variant for hydric site conditions). The use of VNA implies that users of the data **should not expect** that any data could be present.

DNC—Did Not Collect

“DNC” is used to describe variables that should have been collected but were not. There are a number of reasons that data might not have been collected (e.g., staff oversight, equipment failure, safety concerns, environmental conditions, or time constraints). Regardless of the reason data was not collected, if under ideal conditions it should have been, the record in the data entry file reads “DNC.” The use of DNC implies that users **should expect** the data to be present—though it is not.

PNA—Protocol Not Available

The ABMI’s protocols were, and continue to be, implemented in a staged manner. As a result, the collection of many variables began in years subsequent to the start of the prototype or operational phases or where discontinued after a few years of trial. When a variable was not collected because the protocol had yet to be implemented by the ABMI (or was discontinued by the ABMI), the data entry record reads “PNA.” This is a global constraint to the data (i.e., a protocol was not implemented until 2006, therefore, previous years cannot have this variable). PNA is to be used to describe the lack of data collection for entire years.

SNI—Species Not Identified

In various fields related to species identification, “SNI” is used to indicate that the organism was not identified. Some possible reasons that identification was not possible include insufficient or deficient sample collected and lack of field time.

SNR—Species Needs Review

In various fields related to species identification, “SNR” is used to indicate that an organism’s identification needs to be reviewed. Some possible reasons that identification is reviewed include occurrence outside of the organism’s known range and habitat, or no known occurrence within the province.

Terrestrial T01A Site Physical Characteristics Raw Data Metadata

Rotation

Unique ID	T01A_IC00001
Attribute Definition	Reference describing when data was collected at a broad level (i.e., Prototype, Rotation 1, Rotation 2). The rotation number indicates the visit number (i.e. 1 = first visit, 2 = second visit or first revisit).
Value Type	Text
Code	Prototype = 2003--2006 Rotation 1 = 2007 onward Rotation 2 = 2015 onward

ABMI Site

Unique ID	T01A_IC00003
Attribute Definition	Reference number given to each ABMI data collection site. An alphabetized suffix distinguishes a new site from the old site(s). Off grid data collection sites are appended with an OG prefix, 2-5 letter project code prefix, and a 1-2 digit suffix. Short term data collection sites are appended with an ST prefix and a 1-2 digit suffix.
Value Type	Number
Format	1-4 digits & 1 letter (if necessary); OG & 2-5 letter prefix, 1-4 digits, and 1-2 digit suffix
Range	1-1656; OG & 2-5 letter prefix (if applicable)

Year

Unique ID	T01A_IC00002
Attribute Definition	The calendar year representing the annual field program and funding the data are associated with.
Value Type	Date
Format	YYYY

Field Date

Unique ID	T01A_IC00015
Attribute Definition	Day, month, and year data were collected.
Value Type	Date
Format	DD-Mon-YY
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Field Crew Member(s)

Unique ID	T01A_IC00379
Attribute Definition	Initials of the field technician(s) who collected data.
Value Type	Text
Format	2 or 3 letters (UPPERCASE) and 1 number (if necessary); 1 set of initials or a combination of many
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Nearest Town

Unique ID	T01A_IC00218
Attribute Definition	Nearest town to ABMI site in which field crews have accommodation.
Value Type	Text
Missing Values	DNC = Did Not Collect

Public Latitude

Unique ID	T01A_IC00256
Attribute Definition	Geographic coordinates (Northing) for the segment location.
Value Type	Number
Format	6 or more digits
Unit	degree
Missing Values	DNC = Did Not Collect

Public Longitude

Unique ID	T01A_IC00257
Attribute Definition	Geographic coordinates (Easting) for the segment location.
Value Type	Number
Format	6 or more digits
Unit	degree
Missing Values	DNC = Did Not Collect

Collection Methodology

Unique ID	T01A_IC01141
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Attribute Definition	Method used to collect ecosite data; based on how bird recording data is collected. For the nine Bird Point Count Stations a 150m radius is assessed, while a 50m radius is assessed for sites installed with Autonomous Recording Units (ARUs).
Value Type	Code
Code	1 = Bird Point Count Points (2003-2015) 2 = ARU Points (2015 onward) 3 = Linear Off-Grid ARU Points (2015) 4 = OGC Off-Grid ARU Points (2015)

Point Count Station

Unique ID	T01A_IC00248
Attribute Definition	Location where site suitability information was recorded. There are 9 point count stations established per ABMI site with #1 being site centre and the remainder of the stations arranged around site centre at 300 m intervals.
Value Type	Number
Format	1 digit
Code	1 = point count station 1 2 = point count station 2 3 = point count station 3 4 = point count station 4 5 = point count station 5 6 = point count station 6 7 = point count station 7 8 = point count station 8 9 = point count station 9
Missing Values	VNA = Variable Not Applicable

Subpoint

Unique ID	T01A_IC01009
Attribute Definition	Classification of ecosite types into primary (most common) and secondary.
Value Type	Code
Code	P = Primary S = Secondary

Elevation (metres)

Unique ID	T01A_IC00295
Attribute Definition	Height of the ABMI site above sea level. Obtained from Alberta Sustainable Resource Development Digital Elevation Model (DEM).
Value Type	Number
Unit	metres
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Slope (degrees)

Unique ID	T01A_IC00297
Attribute Definition	Percent slope gradient, measured with a clinometer or compass.
Value Type	Number
Range	0-90
Unit	degree
Missing Values	DNC = Did Not Collect PNA = Protocol Not Available VNA = Variable Not Applicable

Aspect (degrees)

Unique ID	T01A_IC00101
Attribute Definition	Direction of the slope, measured by a compass.
Value Type	Number
Range	0-359
Unit	degree
Missing Values	DNC = Did Not Collect PNA = Protocol Not Available VNA = Variable Not Applicable

Terrestrial T01B Site Suitability Raw Data Metadata

Rotation

Unique ID	T01B_IC00001
Attribute Definition	Reference describing when data was collected at a broad level (i.e., Prototype, Rotation 1, Rotation 2). The rotation number indicates the visit number (i.e. 1 = first visit, 2 = second visit or first revisit).
Value Type	Text
Code	Prototype = 2003--2006 Rotation 1 = 2007 onward Rotation 2 = 2015 onward

ABMI Site

Unique ID	T01B_IC00003
Attribute Definition	Reference number given to each ABMI data collection site. An alphabetized suffix distinguishes a new site from the old site(s). Off grid data collection sites are appended with an OG prefix, 2-5 letter project code prefix, and a 1-2 digit suffix. Short term data collection sites are appended with an ST prefix and a 1-2 digit suffix.
Value Type	Number
Format	1-4 digits & 1 letter (if necessary); OG & 2-5 letter prefix, 1-4 digits, and 1-2 digit suffix
Range	1-1656; OG & 2-5 letter prefix (if applicable)

Year

Unique ID	T01B_IC00002
Attribute Definition	The calendar year representing the annual field program and funding the data are associated with.
Value Type	Date
Format	YYYY

Field Date

Unique ID	T01B_IC00015
Attribute Definition	Day, month, and year data were collected.
Value Type	Date
Format	DD-Mon-YY
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Field Crew Member(s)

Unique ID	T01B_IC00379
Attribute Definition	Initials of the field technician(s) who collected data.
Value Type	Text
Format	2 or 3 letters (UPPERCASE) and 1 number (if necessary); 1 set of initials or a combination of many
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Collection Methodology

Unique ID	T01B_IC01141
Attribute Definition	Method used to collect ecosite data; based on how bird recording data is collected. For the nine Bird Point Count Stations a 150m radius is assessed, while a 50m radius is assessed for sites installed with Autonomous Recording Units (ARUs).
Value Type	Code
Code	1 = Bird Point Count Points (2003-2015) 2 = ARU Points (2015 onward) 3 = Linear Off-Grid ARU Points (2015) 4 = OGC Off-Grid ARU Points (2015)

Point Count Station

Unique ID	T01B_IC00248
Attribute Definition	Location where site suitability information was recorded. Data from 2003-2014, there are 9 point count stations established per ABMI site with #1 being site centre and the remainder of the stations arranged around site centre at 300 m intervals. Data from 2015 onwards, the number and location of recordings made depends on the collection methodology.
Value Type	Number
Format	1 digit
Code	1 = point count station 1 2 = point count station 2 3 = point count station 3 4 = point count station 4 5 = point count station 5 6 = point count station 6 7 = point count station 7 8 = point count station 8 9 = point count station 9
Missing Values	VNA = Variable Not Applicable

Subpoint

Unique ID	T01B_IC01009
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Attribute Definition	Classification of ecosite types into primary (most common) and secondary.
Value Type	Code
Code	P = Primary S = Secondary

Canopy Layer

Unique ID	T01B_IC01341
Attribute Definition	Canopy layer of trees (veteran, dominant, and suppressed) and shrubs (>1.3 m, <1.3 m). See the field "Crown Class" under Medium Trees/Snags/Stumps for definition of veteran, dominant, and suppressed trees.
Value Type	Code
Code	Veteran Dominant Suppressed Predominant shrub >1.3 m Predominant shrub < 1.3 m

Common Name

Unique ID	T01B_IC00122
Attribute Definition	Common name of trees and shrubs.
Value Type	Text
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable NONE = Absent from Survey Area PNA = Protocol Not Available SNI = Species Not Identified SNR = Species Needs Review UID = Unable to Identify

Scientific Name

Unique ID	T01B_IC00286
Attribute Definition	Scientific name of trees and shrubs.
Value Type	Text
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable NONE = Absent from Survey Area PNA = Protocol Not Available SNI = Species Not Identified SNR = Species Needs Review UID = Unable to Identify

Taxonomic Resolution

Unique ID	T01B_IC00310
Attribute Definition	Resolution to which trees and shrubs were identified (e.g. Family, Genus, Species etc.).
Value Type	Text

Missing Values VNA = Variable Not Applicable

Unique Taxonomic Identification Number

Unique ID T01B_IC00357
 Attribute Definition Globally unique identifier of trees and shrubs. Unique taxonomic identifiers are generally taken from the International Taxonomic Information System (ITIS; <http://www.itis.gov/>).
 Value Type Number
 Missing Values VNA = Variable Not Applicable

Percent Composition of Tree Species (Dominant Strata)

Unique ID T01B_IC01342
 Attribute Definition Percent composition of the tree species in 5% increments.
 Value Type Text
 Format 1 to 3 digits
 Missing Values DNC = Did Not Collect | VNA = Variable Not Applicable

Rank in Canopy

Unique ID T01B_IC00264
 Attribute Definition For each canopy layer (e.g. dominant trees), it is the rank abundance of species for that layer. [2003-2006 only]
 Value Type Code
 Format 1 digit
 Code 1 = most abundant species | 2 = second most abundant species | 3 = third most abundant species | 4 = fourth most abundant species
 Missing Values DNC = Did Not Collect | VNA = Variable Not Applicable | NONE = Absent from Site | PNA = Protocol Not Available

Average Distance (metres) between Trees

Unique ID T01B_IC01073
 Attribute Definition Estimation of distance (in metres) between trees within a canopy layer (e.g. distance between veteran trees) in the dominant stand type.
 Value Type Number
 Range <0.5 and up, not 0
 Unit metre

Missing Values DNC = Did Not Collect | VNA = Variable Not Applicable

Height of Trees (metres)

Unique ID T01B_IC00185
Attribute Definition Estimate of height (in metres) within 5 general categories for a particular tree.
Value Type Number
Code <5 = less than 5 m in height | 5-10 m = between 5 and 10 m in height | 10-15 = between 10 and 15 m in height | 15-20 = between 15 to 20 m in height | 20-25 = between 20 and 25 m in height | >25 = greater than 25 m in height
Missing Values DNC = Did Not Collect | VNA = Variable Not Applicable

Percent Cover (Shrubs Only)

Unique ID T01B_IC01074
Attribute Definition Estimate of cover (closure) for shrubs.
Value Type Code
Code 0 = 0% cover | <25 = less than 25% cover | 25-50 = between 25% and 50% cover | 51-75 = between 51 and 75% | >75 = greater than 75% cover
Missing Values DNC = Did Not Collect | PNA = Protocol Not Available | VNA = Variable Not Applicable

Terrestrial T01C Site Capability Raw Data Metadata

Rotation

Unique ID	T01C_IC00001
Attribute Definition	Reference describing when data was collected at a broad level (i.e., Prototype, Rotation 1, Rotation 2). The rotation number indicates the visit number (i.e. 1 = first visit, 2 = second visit or first revisit).
Value Type	Text
Code	Prototype = 2003--2006 Rotation 1 = 2007 onward Rotation 2 = 2015 onward

ABMI Site

Unique ID	T01C_IC00003
Attribute Definition	Reference number given to each ABMI data collection site. An alphabetized suffix distinguishes a new site from the old site(s). Off grid data collection sites are appended with an OG prefix, 2-5 letter project code prefix, and a 1-2 digit suffix. Short term data collection sites are appended with an ST prefix and a 1-2 digit suffix.
Value Type	Number
Format	1-4 digits & 1 letter (if necessary); OG & 2-5 letter prefix, 1-4 digits, and 1-2 digit suffix
Range	1-1656; OG & 2-5 letter prefix (if applicable)

Year

Unique ID	T01C_IC00002
Attribute Definition	The calendar year representing the annual field program and funding the data are associated with.
Value Type	Date
Format	YYYY

Field Date

Unique ID	T01C_IC00015
Attribute Definition	Day, month, and year data were collected.
Value Type	Date
Format	DD-Mon-YY
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Field Crew Member(s)

Unique ID	T01C_IC00379
Attribute Definition	Initials of the field technician(s) who collected data.
Value Type	Text
Format	2 or 3 letters (UPPERCASE) and 1 number (if necessary); 1 set of initials or a combination of many
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Collection Methodology

Unique ID	T01C_IC01141
Attribute Definition	Method used to collect ecosite data; based on how bird recording data is collected. For the nine Bird Point Count Stations a 150m radius is assessed, while a 50m radius is assessed for sites installed with Autonomous Recording Units (ARUs).
Value Type	Code
Code	1 = Bird Point Count Points (2003-2015) 2 = ARU Points (2015 onward) 3 = Linear Off-Grid ARU Points (2015) 4 = OGC Off-Grid ARU Points (2015)

Point Count Station

Unique ID	T01C_IC00248
Attribute Definition	Location where site capability information was recorded. There are 9 point count stations established per ABMI site with #1 being site centre and the remainder of the stations arranged around site centre at 300 m intervals.
Value Type	Number
Format	1 digit
Code	1 = point count station 1 2 = point count station 2 3 = point count station 3 4 = point count station 4 5 = point count station 5 6 = point count station 6 7 = point count station 7 8 = point count station 8 9 = point count station 9
Missing Values	VNA = Variable Not Applicable

Subpoint

Unique ID	T01C_IC01009
Attribute Definition	Classification of ecosite types into primary (most common) and secondary.
Value Type	Code

Code P = Primary | S = Secondary

Time Period

Unique ID T01C_IC01131
Attribute Definition C=Current ecosite | H=Historical ecosite
Value Type Text

Ecosite - Nutrient/Moisture Code

Unique ID T01C_IC00162
Attribute Definition Nutrient and moisture code for the ecological site classification of a vegetated site based on the dominant vegetation community that is present.
Value Type Code

Code	1 - PX = Poor/Xeric - Bearberry, Lichen, Bog Cranberry common at some sites. 2 - PM = Poor/Mesic - Labrador Tea, Feather Moss, Bog Cranberry, Bilberry, and Grouse-berry common at some sites. 3 - MX = Medium/Xeric - Hairy Wild Rye, Bearberry, Canada Buffaloberry, and Feather Moss common at some sites. 4 - MM = Medium/Mesic - Low-bush Cranberry, Canada Buffaloberry, Blueberry, Alder, Rose, Saskatoon, Labrador Tea, Bearberry, Thimbleberry, Bog Cranberry, and Feather Moss may be common at some sites. 5 - MG = Medium/Hygic - Horsetail, Dogwood, Rose, Willow, and Feather Moss common at some sites. 6 - RG = Rich/Hygic - Dogwood, Fern, Feather Moss, Rose, Alder, Bracted Honeysuckle, Devil's Club, Fir common at some sites. 7 - NT = Not Treed. 8 - PD = Bog - Poor/Hydric - Labrador Tea, Peat Moss, Lichen, Bog Cranberry and Cloudberry may also be present (Soil saturated for part or all of the year. Undecomposed organic soil substrate). 9 - MD = Poor Fen - Medium/Hydric - Labrador Tea, Peat Moss, Sedge, Bog Cranberry, Dwarf Birch and Willow may also be present (Soil saturated for part or all of the year. Undecomposed organic soil substrate). 10 - RD (2005-2014) = Rich Fen - Rich/Hydric - Dwarf Birch, Willow, Sedge, Grass, Moss (soil saturated for part or all of the year; includes floating mats of vegetation). 10 - RDp (2015-later) = Rich Fen - Rich/Hydric - Dwarf Birch, Willow, Sedge, Grass, Moss (Soil saturated for part or all of the year; undecomposed organic soil substrate; includes floating mats of vegetation). 10.5 - RDm = Wet Meadow - Rich/Hydric - Dominated by sedge, grass, presence of shrub and trees (e.g. willow). Conductivity <15mS/cm; soil, saturated for part or all of the year. Well decomposed, organic soil substrate. 11 - VD = Marsh - Very Rich/Hydric - Cattail, Rush, Reed. Conductivity <15 mS/cm, sedge and grass may also be present (Water is above the rooting zone for most or all of the year). 12 - SD = Swamp - Trees and shrubs present, poorly developed bryophytes, often with pools of water (Water is above the rooting zone for some of the year, mineral or humified organic soil rather than peaty). 13 - AD = Alkali - Conductivity >15 mS/cm, white salt flats at water's edge, (Water is above the rooting zone for most or all of the year). 14 - OW = Open Water - No trees.
Missing Values	DNC = Did Not Collect PNA = Protocol Not Available VNA = Variable Not Applicable

Ecosite - Tree Species Modifier

Unique ID	T01C_IC00164
Attribute Definition	At every vegetation sampling location, the tree species modifier is used to classify the site if trees are present and is combined with the moisture/nutrient category to give an ecological site classification.
Value Type	Code

Code	01a Pine = Jack Pine + Douglas Fir >80% 02a Pine = Jack Pine + Lodgepole Pine >50% 02b Other = Aspen + White Spruce + Engelmann Spruce + Subalpine Fir + Western White Pine >50% 02c Sb = Black Spruce >50% 03a None = No Trees 03b Pine = Jack Pine + Lodgepole Pine >50% 03c AwMix = Aspen >20% 03d Spruce = White Spruce + Engelmann Spruce + Subalpine Fir >50% 04a Pine = Jack Pine + Lodgepole Pine + Subalpine Fir >50% 04b PjMix = Aspen + White Birch + White Spruce > 20% AND Jack Pine >=20% 04c Aw = Aspen >50% 04d AwMix = Aspen > 20% AND White Spruce + Black Spruce + Lodgepole Pine >20% 04e Spruce = White Spruce >50% 05a Poplar = Balsam Poplar + Aspen >50% 05b Spruce = White Spruce + Engelmann Spruce >50% 05c Sb = Black Spruce >50% 06a Pine = Lodgepole Pine >50% 06b Poplar = Balsam Poplar + Aspen >50% 06c Spruce = White Spruce + Engelmann Spruce + Subalpine Fir >50% 07a Alpine = Elevation above tree line 07b Flood = Site disturbed frequently by flooding 07c Ice = Site disturbed frequently by ice or snow 07d Dry = Site in prairies/parkland and receives little precipitation 07e Geo = Geological features not suitable for tree growth 07f Human = Site disturbed recently by humans 08a Sb = >=10% tree cover (may only be in shrub/ground strata), Black Spruce >50% [historic 08a SbLt values were converted to 08a Sb and 08a SbLt is no longer a valid code] 08b Shrub = <10% tree cover 09a SbLt = >=10% tree cover (may only be in shrub/ground strata), Black Spruce + Tamarack >50% 09b Shrub = <10% tree cover 10a SbLt = >=10% tree cover (may only be in shrub/ground strata), Black Spruce + Tamarack >50% 10b Shrub = <10% tree cover AND >=10% shrub cover 10c None = <10% tree cover AND <10% shrub cover 10.5a Tree = >=10% tree cover (usually along wetland edge; may only be in shrub/ground strata 10.5b Shrub = <10% tree cover AND >=10% shrub cover 10.5c None = <10% tree cover AND <10% shrub cover 11a None = usually along a water body edge >=10% emergent vegetation cover, <10% tree cover 12a Tree = >10% tree cover 12b Shrub = <10% tree cover 13a None = <10% shrub/tree cover 14a Lake = In standing water <10% emergent vegetation cover 14b River = In flowing water <10% emergent vegetation cover
Missing Values	DNC = Did Not Collect PNA = Protocol Not Available VNA = Variable Not Applicable

Ecosite - Structural Stage

Unique ID	T01C_IC00163
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Attribute Definition	Describes the structural stage of the ecosite, as determined after the ecological-site type has been designated i.e. the nutrient/moisture code designation and tree species modifier. The structural stage code starts with a number that defines 1) tree-dominated, 2) non-tree dominated, or 3) open water ecosites. The subsequent letter codes further describe the ecosite structure, in terms of type of vegetation (or dominant substrate type for non-vegetated sites), as well as height and density of vegetation.
Value Type	Code
Code	For full explanation of the code definitions, see Terrestrial/Wetland Metadata Appendix: Detailed Code Definitions. 1) = Tree Dominated Ecosites Tree Height: TS = Short TT = Tall Tree Density: D = Dense S = Sparse Tree Arrangement: C = Complex N = Non-Complex 2) = Non-Tree Dominated Ecosites N = Non-Vegetated Substrate Type: R = Rock S = Sand B = Beach M = Mineral Soil O = Organic Soil G = Ground Vegetation Only: Vegetation Type: B = Bryoid/Lichen F = Forbs G = Graminoid R = Reeds and Rushes (Marsh) Vegetation Density: D = Dense M = Moderate S = Sparse S = Shrubs present Shrub Height: L = Low T = Tall Shrub Density: D = Dense M = Moderate S = Sparse TL = Low trees present Low Tree Density: D = Dense M = Moderate S = Sparse 3) = Open Water Dominated Communities Vegetation Type: OV = Vegetated ON = Non-Vegetated Vegetation Height: S = Short Submerged M = Medium Submerged T = Tall Submerged F = Floating Vegetation Density: D = Dense M = Moderate S = Sparse
Missing Values	DNC = Did Not Collect PNA = Protocol Not Available VNA = Variable Not Applicable

Percent Area of Ecological Site Classification

Unique ID	T01C_IC01343
Attribute Definition	Percent of the area surrounding the ABMI site/point count station described by the primary and secondary ecological site classification (5% increments).
Value Type	Number
Format	2 digits
Range	0-100%
Unit	percent
Missing Values	DNC = Did Not Collect PNA = Protocol Not Available VNA = Variable Not Applicable

Terrestrial T01D Ground Cover Raw Data Metadata

Rotation

Unique ID	T01D_IC00001
Attribute Definition	Reference describing when data was collected at a broad level (i.e., Prototype, Rotation 1, Rotation 2). The rotation number indicates the visit number (i.e. 1 = first visit, 2 = second visit or first revisit).
Value Type	Text
Code	Prototype = 2003--2006 Rotation 1 = 2007 onward Rotation 2 = 2015 onward

ABMI Site

Unique ID	T01D_IC00003
Attribute Definition	Reference number given to each ABMI data collection site. An alphabetized suffix distinguishes a new site from the old site(s). Off grid data collection sites are appended with an OG prefix, 2-5 letter project code prefix, and a 1-2 digit suffix. Short term data collection sites are appended with an ST prefix and a 1-2 digit suffix.
Value Type	Number
Format	1-4 digits & 1 letter (if necessary); OG & 2-5 letter prefix, 1-4 digits, and 1-2 digit suffix
Range	1-1656; OG & 2-5 letter prefix (if applicable)

Year

Unique ID	T01D_IC00002
Attribute Definition	The calendar year representing the annual field program and funding the data are associated with.
Value Type	Date
Format	YYYY

Field Date

Unique ID	T01D_IC00015
Attribute Definition	Day, month, and year data were collected.
Value Type	Date
Format	DD-Mon-YY
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Field Crew Member(s)

Unique ID	T01D_IC00379
Attribute Definition	Initials of the field technician(s) who collected data.
Value Type	Text
Format	2 or 3 letters (UPPERCASE) and 1 number (if necessary); 1 set of initials or a combination of many
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Collection Methodology

Unique ID	T01D_IC01141
Attribute Definition	Method used to collect ecosite data; based on how bird recording data is collected. For the nine Bird Point Count Stations a 150m radius is assessed, while a 50m radius is assessed for sites installed with Autonomous Recording Units (ARUs).
Value Type	Code
Code	1 = Bird Point Count Points (2003-2015) 2 = ARU Points (2015 onward) 3 = Linear Off-Grid ARU Points (2015) 4 = OGC Off-Grid ARU Points (2015)

Point Count Station

Unique ID	T01D_IC00248
Attribute Definition	Location where ground cover information was recorded. There are 9 point count stations established per ABMI site with #1 being site centre and the remainder of the stations arranged around site centre at 300 m intervals.
Value Type	Number
Format	1 digit
Code	1 = point count station 1 2 = point count station 2 3 = point count station 3 4 = point count station 4 5 = point count station 5 6 = point count station 6 7 = point count station 7 8 = point count station 8 9 = point count station 9
Missing Values	VNA = Variable Not Applicable

Subpoint

Unique ID	T01D_IC01009
Attribute Definition	Classification of ecosite types into primary (most common) and secondary.
Value Type	Code

Code P = Primary | S = Secondary

Percent Cover of Standing Dead Herbaceous

Unique ID	T01D_IC01033
Attribute Definition	Cover estimate of standing dead herbaceous cover in 4 general categories that would be obtained if a photograph had been taken from a height of 2.0 m. [2003-2016]
Value Type	Number
Code	0 = no standing dead herbaceous vegetation <25 = less than 25% cover of standing dead herbaceous vegetation 25-50 = between 25% and 50% cover of standing dead herbaceous vegetation 51-75 = between 51 and 75% cover of standing dead herbaceous vegetation >75 = greater than 75% cover of standing dead herbaceous vegetation.
Missing Values	DNC = Did Not Collect PNA = Protocol Not Available VNA = Variable Not Applicable

Height of Standing Dead Herbaceous (metres)

Unique ID	T01D_IC00183
Attribute Definition	Estimate of height (in metres) of standing dead herbaceous cover in 0.1 m increments. [2003-2016]
Value Type	Number
Unit	metre
Missing Values	DNC = Did Not Collect PNA = Protocol Not Available VNA = Variable Not Applicable

Predominant Type of Live Herbaceous

Unique ID	T01D_IC00250
Attribute Definition	Low vegetation, <0.5 m tall, that is or will be most abundant in the quadrant; listed in general categories.
Value Type	Code
Code	Grass Herb Lichen Moss Sedge Shrub Weed
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable NONE = Absent from site

Percent Cover of Bare Ground

Unique ID	T01D_IC01036
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Attribute Definition	Percent cover estimate of bare ground in the quadrant in 5% increments.
Value Type	Code
Code	0 <1 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Percent Cover of Water

Unique ID	T01D_IC01037
Attribute Definition	Percent cover estimate of water covering the quadrant in 5% increments.
Value Type	Code
Code	0 <1 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Terrestrial T01E Site Disturbance Raw Data Metadata

Rotation

Unique ID	T01E_IC00001
Attribute Definition	Reference describing when data was collected at a broad level (i.e., Prototype, Rotation 1, Rotation 2). The rotation number indicates the visit number (i.e. 1 = first visit, 2 = second visit or first revisit).
Value Type	Text
Code	Prototype = 2003--2006 Rotation 1 = 2007 onward Rotation 2 = 2015 onward

ABMI Site

Unique ID	T01E_IC00003
Attribute Definition	Reference number given to each ABMI data collection site. An alphabetized suffix distinguishes a new site from the old site(s). Off grid data collection sites are appended with an OG prefix, 2-5 letter project code prefix, and a 1-2 digit suffix. Short term data collection sites are appended with an ST prefix and a 1-2 digit suffix.
Value Type	Number
Format	1-4 digits & 1 letter (if necessary); OG & 2-5 letter prefix, 1-4 digits, and 1-2 digit suffix
Range	1-1656; OG & 2-5 letter prefix (if applicable)

Year

Unique ID	T01E_IC00002
Attribute Definition	The calendar year representing the annual field program and funding the data are associated with.
Value Type	Date
Format	YYYY

Field Date

Unique ID	T01E_IC00015
Attribute Definition	Day, month, and year data were collected.
Value Type	Date
Format	DD-Mon-YY
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Field Crew Member(s)

Unique ID	T01E_IC00379
Attribute Definition	Initials of the field technician(s) who collected data.
Value Type	Text
Format	2 or 3 letters (UPPERCASE) and 1 number (if necessary); 1 set of initials or a combination of many
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Collection Methodology

Unique ID	T01E_IC01141
Attribute Definition	Method used to collect ecosite data; based on how bird recording data is collected. For the nine Bird Point Count Stations a 150m radius is assessed, while a 50m radius is assessed for sites installed with Autonomous Recording Units (ARUs).
Value Type	Code
Code	1 = Bird Point Count Points (2003-2015) 2 = ARU Points (2015 onward) 3 = Linear Off-Grid ARU Points (2015) 4 = OGC Off-Grid ARU Points (2015)

Point Count Station

Unique ID	T01E_IC00248
Attribute Definition	Location where site capability information was recorded. There are 9 point count stations established per ABMI site with #1 being site centre and the remainder of the stations arranged around site centre at 300 m intervals.
Value Type	Number
Format	1 digit
Code	1 = point count station 1 2 = point count station 2 3 = point count station 3 4 = point count station 4 5 = point count station 5 6 = point count station 6 7 = point count station 7 8 = point count station 8 9 = point count station 9
Missing Values	VNA = Variable Not Applicable

Subpoint

Unique ID	T01E_IC01009
Attribute Definition	Classification of ecosite types into primary (most common) and secondary.
Value Type	Code

Code P = Primary | S = Secondary

Human or Natural Disturbance

Unique ID T01E_IC00997
Attribute Definition Disturbance can be due to human activity or natural processes.
Value Type Code
Code Human | Natural
Missing Values DNC = Did Not Collect | VNA = Variable Not Applicable

Disturbance Type

Unique ID T01E_IC00994
Attribute Definition Evidence of human-caused or natural stand disturbances; note all minor to extensive disturbance caused by humans (logging debris or stumps, agricultural/livestock activity, seismic lines, roads, trails etc) or naturally caused (e.g. fire scars, insect defoliation, wind caused mortality/blow down).
Value Type Code

Code	BARE = Bare - Human caused bare ground for which the cause cannot be determined CATTLE [2011-2012] = Cattle CULT = Cultivated Crop/Field - Any type of cultivated field that is used to grow agriculture crops HARV = Harvest - Any type of forest harvesting (i.e., clear-cut, partial-cut, understory retention, etc.) <30 years old) IND = Industrial - Any type of building, roadway, yard, etc associated with industrial development PAST = Pasture - Any type of pasture (tame or native), grazing reserve, etc. Linear = Any type of right-of-way, cutline, seismic, pipeline, powerline, etc. [2003-2009] PIPE = Linear - Pipeline POWER = Linear - Powerline RAIL = Linear - Rail line RES = Residential Rural - Any type of human dwelling, farm building, or farm yard in a rural or acreage setting ROAD = Linear - Road - Any type of road for which the type of surface improvement is unknown ROADG = Linear - Road Gravel - Any type of road with an unpaved but improved (i.e. gravel) surface ROADP = Linear - Road Paved - Any type of road with paved surface SEIS = Linear - Seismic or Cutline of any type TRAIL = Linear - Road Other - Any type of truck or ATV trail with an unimproved surface URB = Urban - Any type of human dwelling, associated building, or yard/driveway/road in an urban setting WELL = Well Pad - Any type of area cleared for oil/gas/CBM pump jacks or well heads BEAV = Beaver - Any evidence of beaver activity altering the landscape or vegetation DISEASE = Disease - Any evidence of vegetation experiencing disease outbreak EROS = Erosion - Evidence of wearing away of soil by precipitation or wind FIRE = Fire - Any evidence of fire damage, scarring or burning (<30 years old) FLOOD = Flood - Evidence of flooding including high water mark, dead trees, etc. ICE = Snow/Ice - Evidence of vegetation breakage caused by snow or ice INSECT = Insect - Any evidence of vegetation experiencing insect attack WIND = Wind Throw - Evidence of wind damage including snapped or uprooted trees OTHER = Other disturbance UNKN = Unknown disturbance NONE = No disturbance
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Disturbance Type (%)

Unique ID	T01E_IC00996
Attribute Definition	% of area disturbed
Value Type	Code
Code	0 < 1 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100
Missing Values	DNC = Did Not Collect PNA = Protocol Not Available VNA = Variable Not Applicable

Terrestrial T02A Surface Substrate Raw Data Metadata

Rotation

Unique ID	T02A_IC00001
Attribute Definition	Reference describing when data was collected at a broad level (i.e., Prototype, Rotation 1, Rotation 2). The rotation number indicates the visit number (i.e. 1 = first visit, 2 = second visit or first revisit).
Value Type	Text
Code	Prototype = 2003--2006 Rotation 1 = 2007 onward Rotation 2 = 2015 onward

ABMI Site

Unique ID	T02A_IC00003
Attribute Definition	Reference number given to each ABMI data collection site. An alphabetized suffix distinguishes a new site from the old site(s). Off grid data collection sites are appended with an OG prefix, 2-5 letter project code prefix, and a 1-2 digit suffix. Short term data collection sites are appended with an ST prefix and a 1-2 digit suffix.
Value Type	Number
Format	1-4 digits & 1 letter (if necessary); OG & 2-5 letter prefix, 1-4 digits, and 1-2 digit suffix
Range	1-1656; OG & 2-5 letter prefix (if applicable)

Year

Unique ID	T02A_IC00002
Attribute Definition	The calendar year representing the annual field program and funding the data are associated with.
Value Type	Date
Format	YYYY

Field Date

Unique ID	T02A_IC00015
Attribute Definition	Day, month, and year data were collected.
Value Type	Date
Format	DD-Mon-YY
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Field Crew Member(s)

Unique ID	T02A_IC00379
Attribute Definition	Initials of the field technician(s) who collected data.
Value Type	Text
Format	2 or 3 letters (UPPERCASE) and 1 number (if necessary); 1 set of initials or a combination of many
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Transect (North/South)

Unique ID	T02A_IC00340
Attribute Definition	Indicates the transect (north or south) where the surface substrate was sampled.
Value Type	Code
Format	1 letter (UPPERCASE)
Code	N = North S = South
Missing Values	DNC = Did Not Collect

Sample Position (metres)

Unique ID	T02A_IC00284
Attribute Definition	Location along the 30 metre transect where surface substrate was measured. In lowland areas (i.e., black spruce bogs) we recorded depth every 4 metres along each transect rather than every 2 metres.
Value Type	Code
Format	1-2 digits
Code	2 4 6 8 10 12 14 16 18 20 22 24 26 28 30

Organic Depth (centimetres)

Unique ID	T02A_IC00224
Attribute Definition	Depth (in centimetres) of the organic layer of soil, taken in 0.5 cm increments. We defined organic matter as the litter, fermented, and humus (LFH) layer of the soil horizon. Variations in methodology apply.
Value Type	Number
Format	1-4 digits
Range	<1, 0-500
Unit	centimetre

Missing Values DNC = Did Not Collect | VNA = Variable Not Applicable

Buried Wood Depth (centimetres)

Unique ID	T02A_IC00114
Attribute Definition	Depth (in centimetres) of wood in all decay stages found >50% below the surface. Wood that has organic material growing over it (i.e. decay stage 5) and is <10 centimetres thick is considered organics.
Value Type	Number
Format	2-4 digits
Range	0.0-500.0
Unit	centimetre
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Description

Unique ID	T02A_IC01077
Attribute Definition	Description of reason for partial sample.
Value Type	Text
Format	1-7 letters (Uppercase)
Code	B = Bedrock B-S = Bedrock - Side Sample D = Large Down Woody Material (DWM) D-S = Large DWM - Side Sample HF = Hit Frost HO = Hit Object M = Mineral Soil M-S = Mineral Soil - Side Sample R = Rocks, Cobbles and Stones R-S = Rock - Side Sample W = Water W-S = Water - Side Sample None = Full Sample Taken LFH>40 [2010 only] = LFH layer is greater than 40cm deep, measurements taken every 4 metres
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Slope

Unique ID	T02A_IC01038
Attribute Definition	Position of the transect with respect to the slope.
Value Type	Code

Code	C = Crest - situated in a relatively level area on the top of a hill S1 = Slope - situated on the side of a hill, a modifier of 1 indicates slopes of 2-5 degrees, S2 = Slope - situated on the side of a hill, a modifier of 2 indicates slopes of 6-10 degrees, S3 = Slope - situated on the side of a hill, a modifier of 3 indicates slopes of 11-30 degrees, S4 = Slope - situated on the side of a hill, a modifier of 4 indicates slopes of >30 degrees, T = Toe - situated at the bottom of a hill where the ground surface transitions from a slope to level L = Level - situated in an area with <2 degrees slope D = Depression - situated in an area that accumulates water after rains
Missing Values	DNC = Did Not Collect PNA = Protocol Not Available VNA = Variable Not Applicable

Aspect (degrees)

Unique ID	T02A_IC01039
Attribute Definition	Direction of the slope, measured by a compass.
Value Type	Number
Range	0-359
Unit	degree
Missing Values	DNC = Did Not Collect PNA = Protocol Not Available VNA = Variable Not Applicable

Terrestrial T02B Surface Substrate Disturbance Raw Data Metadata

Rotation

Unique ID	T02B_IC00001
Attribute Definition	Reference describing when data was collected at a broad level (i.e., Prototype, Rotation 1, Rotation 2). The rotation number indicates the visit number (i.e. 1 = first visit, 2 = second visit or first revisit).
Value Type	Text
Code	Prototype = 2003--2006 Rotation 1 = 2007 onward Rotation 2 = 2015 onward

ABMI Site

Unique ID	T02B_IC00003
Attribute Definition	Reference number given to each ABMI data collection site. An alphabetized suffix distinguishes a new site from the old site(s). Off grid data collection sites are appended with an OG prefix, 2-5 letter project code prefix, and a 1-2 digit suffix. Short term data collection sites are appended with an ST prefix and a 1-2 digit suffix.
Value Type	Number
Format	1-4 digits & 1 letter (if necessary); OG & 2-5 letter prefix, 1-4 digits, and 1-2 digit suffix
Range	1-1656; OG & 2-5 letter prefix (if applicable)

Year

Unique ID	T02B_IC00002
Attribute Definition	The calendar year representing the annual field program and funding the data are associated with.
Value Type	Date
Format	YYYY

Field Date

Unique ID	T02B_IC00015
Attribute Definition	Day, month, and year data were collected.
Value Type	Date
Format	DD-Mon-YY

Missing Values DNC = Did Not Collect | VNA = Variable Not Applicable

Field Crew Member(s)

Unique ID T02B_IC00379
 Attribute Definition Initials of the field technician(s) who collected data.
 Value Type Text
 Format 2 or 3 letters (UPPERCASE) and 1 number (if necessary); 1 set of initials or a combination of many
 Missing Values DNC = Did Not Collect | VNA = Variable Not Applicable

Transect (North/South)

Unique ID T02B_IC00340
 Attribute Definition Indicates the transect (north or south) where the surface substrate was sampled.
 Value Type Code
 Format 1 letter (UPPERCASE)
 Code N = North | S = South
 Missing Values DNC = Did Not Collect

Human or Natural Disturbance

Unique ID T02B_IC00997
 Attribute Definition Disturbance can be due to human activity or natural processes.
 Value Type Code
 Code Human | Natural
 Missing Values DNC = Did Not Collect | VNA = Variable Not Applicable

Disturbance Type

Unique ID T02B_IC00994
 Attribute Definition Evidence of human-caused or natural stand disturbances; note all minor to extensive disturbance caused by humans (logging debris or stumps, agricultural/livestock activity, seismic lines, roads, trails etc) or naturally caused (e.g. fire scars, insect defoliation, wind caused mortality/blow down).
 Value Type Code

Code	BARE = Bare - Human caused bare ground for which the cause cannot be determined CATTLE [2011-2012] = Cattle CULT = Cultivated Crop/Field - Any type of cultivated field that is used to grow agriculture crops HARV = Harvest - Any type of forest harvesting (i.e., clear-cut, partial-cut, understory retention, etc.) <30 years old) IND = Industrial - Any type of building, roadway, yard, etc associated with industrial development PAST = Pasture - Any type of pasture (tame or native), grazing reserve, etc. Linear = Any type of right-of-way, cutline, seismic, pipeline, powerline, etc. [2003-2009] PIPE = Linear - Pipeline POWER = Linear - Powerline RAIL = Linear - Rail line RES = Residential Rural - Any type of human dwelling, farm building, or farm yard in a rural or acreage setting ROAD = Linear - Road - Any type of road for which the type of surface improvement is unknown ROADG = Linear - Road Gravel - Any type of road with an unpaved but improved (i.e. gravel) surface ROADP = Linear - Road Paved - Any type of road with paved surface SEIS = Linear - Seismic or Cutline of any type TRAIL = Linear - Road Other - Any type of truck or ATV trail with an unimproved surface URB = Urban - Any type of human dwelling, associated building, or yard/driveway/road in an urban setting WELL = Well Pad - Any type of area cleared for oil/gas/CBM pump jacks or well heads BEAV = Beaver - Any evidence of beaver activity altering the landscape or vegetation DISEASE = Disease - Any evidence of vegetation experiencing disease outbreak EROS = Erosion - Evidence of wearing away of soil by precipitation or wind FIRE = Fire - Any evidence of fire damage, scarring or burning (<30 years old) FLOOD = Flood - Evidence of flooding including high water mark, dead trees, etc. ICE = Snow/Ice - Evidence of vegetation breakage caused by snow or ice INSECT = Insect - Any evidence of vegetation experiencing insect attack WIND = Wind Throw - Evidence of wind damage including snapped or uprooted trees OTHER = Other disturbance UNKN = Unknown disturbance NONE = No disturbance
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Disturbance Type (%)

Unique ID	T02B_IC01175
Attribute Definition	% of area disturbed
Value Type	Code
Code	< 1 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80 85 90 95 100
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Terrestrial T09 Trees, Snags and Stumps Raw Data Metadata

Rotation

Unique ID	T09_IC00001
Attribute Definition	Reference describing when data was collected at a broad level (i.e., Prototype, Rotation 1, Rotation 2). The rotation number indicates the visit number (i.e. 1 = first visit, 2 = second visit or first revisit).
Value Type	Text
Code	Prototype = 2003--2006 Rotation 1 = 2007 onward Rotation 2 = 2015 onward

ABMI Site

Unique ID	T09_IC00003
Attribute Definition	Reference number given to each ABMI data collection site. An alphabetized suffix distinguishes a new site from the old site(s). Off grid data collection sites are appended with an OG prefix, 2-5 letter project code prefix, and a 1-2 digit suffix. Short term data collection sites are appended with an ST prefix and a 1-2 digit suffix.
Value Type	Number
Format	1-4 digits & 1 letter (if necessary); OG & 2-5 letter prefix, 1-4 digits, and 1-2 digit suffix
Range	1-1656; OG & 2-5 letter prefix (if applicable)

Year

Unique ID	T09_IC00002
Attribute Definition	The calendar year representing the annual field program and funding the data are associated with.
Value Type	Date
Format	YYYY

Field Date

Unique ID	T09_IC00015
Attribute Definition	Day, month, and year data were collected.
Value Type	Date
Format	DD-Mon-YY
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Field Crew Member(s)

Unique ID	T09_IC00379
Attribute Definition	Initials of the field technician(s) who collected data.
Value Type	Text
Format	2 or 3 letters (UPPERCASE) and 1 number (if necessary); 1 set of initials or a combination of many
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Quadrant

Unique ID	T09_IC00261
Attribute Definition	Quadrant of the site that trees, snags, and stumps were measured.
Value Type	Code
Format	2-letter code (UPPERCASE)
Code	NE = Northeast Quadrant SE = Southeast Quadrant SW = Southwest Quadrant NW = Northwest Quadrant N = Northeast or Northwest Quadrant S = Southeast or Southwest Quadrant
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Plot

Unique ID	T09_IC00244
Attribute Definition	Indicates which nested plot that was used to measure trees, snags, and stumps. Small trees/snags measured in 5 x 5 metre plot; medium-large trees/snags/stumps measured in 10 x 10 metre plot; large trees/snags/stumps measured in 25 x 25 metre plot.
Value Type	Code
Format	1-letter code (UPPERCASE)
Code	S = Small Plot (5 x 5 m) M = Medium Plot (10 x 10 m) L = Large Plot (25 x 25 m), 2003-2006 25 x 50 m
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Plot Area (metres squared)

Unique ID	T09_IC00245
Attribute Definition	Indicates the plot area in which trees, snags, and stumps were measured; area calculated from the size of the tree/snag/stump plots and expressed in square metres.

Value Type	Code
Code	25 100 625 1250 [2003-2006]
Missing Values	VNA = Variable Not Applicable

Common Name

Unique ID	T09_IC00122
Attribute Definition	Common name of tree, snag, or stump species encountered within the plot.
Value Type	Text
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable NONE = Absent from Survey Area PNA = Protocol Not Available SNI = Species Not Identified SNR = Species Needs Review UID = Unable to Identify

Scientific Name

Unique ID	T09_IC00286
Attribute Definition	Scientific name of tree, snag, or stump species encountered within the plot.
Value Type	Text
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable NONE = Absent from Survey Area PNA = Protocol Not Available SNI = Species Not Identified SNR = Species Needs Review UID = Unable to Identify

Taxonomic Resolution

Unique ID	T09_IC00310
Attribute Definition	Resolution to which tree, snag, or stump was identified (e.g. Family, Genus, Species etc.).
Value Type	Text
Missing Values	VNA = Variable Not Applicable

Unique Taxonomic Identification Number

Unique ID	T09_IC00357
Attribute Definition	Globally unique identifier of tree, snag, or stump species encountered within the plot. Unique taxonomic identifiers are generally taken from the International Taxonomic Information System (ITIS; http://www.itis.gov/) .
Value Type	Number

Missing Values VNA = Variable Not Applicable

Condition

Unique ID T09_IC00124
 Attribute Definition Classification of encountered tree species as Alive, Dead (snag >=1.3 metres high), or Stump (<1.3 metres high).
 Value Type Code
 Code Alive | Dead | Stump
 Missing Values DNC = Did Not Collect | VNA = Variable Not Applicable

DBH (centimetres)

Unique ID T09_IC00143
 Attribute Definition Diameter of tree or snag at breast height (1.3 metre) and inside Diameter (no bark) of stumps. DBH is measured in 0.1 cm increments for trees/snags/stumps <= 7 cm in DBH and in 0.5 cm increments for trees/snags/stumps >7 cm in DBH. Prior to 2012, increments of 0.1 cm were used for all DBH measurements.
 Value Type Number
 Unit centimetre
 Missing Values DNC = Did Not Collect | VNA = Variable Not Applicable

Decay Class

Unique ID T09_IC00145
 Attribute Definition Decay classification given to snags (dead trees), and stumps based on 6 stages of decay.
 Value Type Code
 Code 1 = Recently killed, all twigs/branches present, wood hard, bark (normally) intact. | 2 = Twigs and small branches missing (major branches remain), wood hard. | 3 = No branches, bole mostly intact, wood starting to soften. | 1-2S = Recently killed, wood hard, bark (normally) intact. | 3S = Wood starting to soften. | 4S = Wood soft throughout the snag.
 Missing Values DNC = Did Not Collect | VNA = Variable Not Applicable

Top Height (metres)

Unique ID T09_IC00319

Attribute Definition	Height of tree or snag to the top, and stumps in 0.1 metre increments.
Value Type	Number
Range	>/ = 0.1 m
Unit	metre
Missing Values	DNC = Did Not Collect PNA = Protocol Not Available VNA = Variable Not Applicable

Base Height (metres)

Unique ID	T09_IC00111
Attribute Definition	Height to the crown base of tree, in 0.1 metre increments.
Value Type	Number
Range	>/ = 0.1 m
Unit	metre
Missing Values	DNC = Did Not Collect PNA = Protocol Not Available VNA = Variable Not Applicable

Height Measured/Estimated

Unique ID	T09_IC00182
Attribute Definition	Indicates whether field staff measured or estimated the top heights of all applicable trees and base heights of medium to large sized trees if sub-sampling occurred.
Value Type	Code
Code	M = Measured E = Estimated
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Crown Class

Unique ID	T09_IC00134
Attribute Definition	Tree classification in relation to the immediately surrounding trees.
Value Type	Code

Code	V = Veteran - Trees that are considerably older than rest of the stand, usually remaining from a previous forest D = Dominant - Trees with well developed crowns extending slightly above the general level of surrounding trees, receiving full light from above and partly from the side. C = Co-dominant - Trees with crowns (slightly smaller than dominant and crowded from the sides) forming the general level of surrounding trees, receiving full light from above and little from the side. I = Intermediate - Trees with crowns (usually small and quite crowded) below, but extending to the general level of surrounding trees, receiving little light from above and none from the sides. S = Suppressed - Trees with crowns entirely below the general level of the surrounding trees, receiving virturally no direct light from above or the side.
Missing Values	DNC = Did Not Collect PNA = Protocol Not Available VNA = Variable Not Applicable

Terrestrial T10A Tree Core Ages (2007-2015) Raw Data Metadata

Rotation

Unique ID	T10A_IC00001
Attribute Definition	Reference describing when data was collected at a broad level (i.e., Prototype, Rotation 1, Rotation 2). The rotation number indicates the visit number (i.e. 1 = first visit, 2 = second visit or first revisit).
Value Type	Text
Code	Prototype = 2003--2006 Rotation 1 = 2007 onward Rotation 2 = 2015 onward

ABMI Site

Unique ID	T10A_IC00003
Attribute Definition	Reference number given to each ABMI data collection site. An alphabetized suffix distinguishes a new site from the old site(s). Off grid data collection sites are appended with an OG prefix, 2-5 letter project code prefix, and a 1-2 digit suffix. Short term data collection sites are appended with an ST prefix and a 1-2 digit suffix.
Value Type	Number
Format	1-4 digits & 1 letter (if necessary); OG & 2-5 letter prefix, 1-4 digits, and 1-2 digit suffix
Range	1-1656; OG & 2-5 letter prefix (if applicable)

Year

Unique ID	T10A_IC00002
Attribute Definition	The calendar year representing the annual field program and funding the data are associated with.
Value Type	Date
Format	YYYY

Field Date

Unique ID	T10A_IC00015
Attribute Definition	Day, month, and year data were collected.
Value Type	Date
Format	DD-Mon-YY
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Field Crew Member(s)

Unique ID	T10A_IC00379
Attribute Definition	Initials of the field technician(s) who collected data.
Value Type	Text
Format	2 or 3 letters (UPPERCASE) and 1 number (if necessary); 1 set of initials or a combination of many
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Identification Date

Unique ID	T10A_IC00192
Attribute Definition	Day, month, and year data was identified by technician/specialist.
Value Type	Date
Format	DD-Mon-YY
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Identification Analyst

Unique ID	T10A_IC00012
Attribute Definition	Initials for the technicians/specialists identifying the specimens.
Value Type	Code
Format	2-3 letter code (UPPERCASE) and 1 digit if necessary
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Quadrant

Unique ID	T10A_IC00262
Attribute Definition	Quadrant where core or cookie was collected from tree species to measure age.
Value Type	Code
Code	1 ha = 1 hectare plot NE = Northeast Quadrant SE = Southeast Quadrant SW = Southwest Quadrant NW = Northwest Quadrant N = Northern Quadrants S = Southern Quadrants

Tree Core Type

Unique ID	T10A_IC00349
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Attribute Definition	Tree cores taken for: Largest Tree i.e. tree with largest DBH found within the central 1 ha of the ABMI site; Leading Species i.e. dominant or co-dominant trees that make up the majority of the stand composition (most abundant species), not including veterans found within each of the four quadrants, and; Secondary Species i.e. dominant or co-dominant trees that make up the second most abundant species but must be $\geq 20\%$ of the stand composition, not including veterans found in each of the four quadrants.
Value Type	Code
Code	Largest Tree Leading Species Secondary Species

Sample Type

Unique ID	T10A_IC00285
Attribute Definition	Indicates whether a core or cookie type of sample was taken
Value Type	Code
Code	Cookie Core
Missing Values	DNC = Did Not Collect PNA = Protocol Not Available VNA = Variable Not Applicable

Common Name

Unique ID	T10A_IC00122
Attribute Definition	Common name of tree from which a core or cookie was taken.
Value Type	Text
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable NONE = Absent from Survey Area PNA = Protocol Not Available SNI = Species Not Identified SNR = Species Needs Review UID = Unable to Identify

Scientific Name

Unique ID	T10A_IC00286
Attribute Definition	Scientific name of tree from which a core or cookie was taken.
Value Type	Text
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable NONE = Absent from Survey Area PNA = Protocol Not Available SNI = Species Not Identified SNR = Species Needs Review UID = Unable to Identify

Taxonomic Resolution

Unique ID	T10A_IC00310
Attribute Definition	Resolution to which taxa was identified (e.g. Family, Genus, Species etc.).
Value Type	Text
Missing Values	VNA = Variable Not Applicable

Unique Taxonomic Identification Number

Unique ID	T10A_IC00357
Attribute Definition	Globally unique identifier of tree from which a core or cookie was taken. Unique taxonomic identifiers are generally taken from the International Taxonomic Information System (ITIS; http://www.itis.gov/).
Value Type	Number
Missing Values	VNA = Variable Not Applicable

Tree Height (metres)

Unique ID	T10A_IC00350
Attribute Definition	Height of the tree in metres, to the top leaf or stem (0.1 m increments).
Value Type	Number
Format	1 or more digits
Range	1.3 and up
Unit	metre
Missing Values	DNC = Did Not Collect PNA = Protocol Not Available VNA = Variable Not Applicable

DBH (centimetres)

Unique ID	T10A_IC00144
Attribute Definition	Diameter of tree at breast height (1.3 metre) for the largest tree, leading species, or secondary species from which a tree core or cookie was collected to measure age
Value Type	Number
Unit	centimetre
Missing Values	DNC = Did Not Collect PNA = Protocol Not Available VNA = Variable Not Applicable

Age (years)

Unique ID	T10A_IC00097
Attribute Definition	Unadjusted number of tree rings counted from cores or cookies, taken at 1.3 m (DBH) height.
Value Type	Number
Format	1 or more digits
Range	1 and up
Unit	years
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Age Estimation Confidence

Unique ID	T10A_IC01119
Attribute Definition	This field describes the confidence of the tree age estimation
Value Type	Code
Code	1 = Excellent: Age is accurate to +/-1 2 = Good: Age is accurate to +/-2 3 = Fair: Age is accurate to +/-3 4 = Poor: Age is accurate to +/-4 5 = Very Poor: Age is very inaccurate and variable (age correction is >20 yrs)
Missing Values	DNC = Did Not Collect

Terrestrial T10B Tree Damages Raw Data Metadata

Rotation

Unique ID	T10B_IC00001
Attribute Definition	Reference describing when data was collected at a broad level (i.e., Prototype, Rotation 1, Rotation 2). The rotation number indicates the visit number (i.e. 1 = first visit, 2 = second visit or first revisit).
Value Type	Text
Code	Prototype = 2003--2006 Rotation 1 = 2007 onward Rotation 2 = 2015 onward

ABMI Site

Unique ID	T10B_IC00003
Attribute Definition	Reference number given to each ABMI data collection site. An alphabetized suffix distinguishes a new site from the old site(s). Off grid data collection sites are appended with an OG prefix, 2-5 letter project code prefix, and a 1-2 digit suffix. Short term data collection sites are appended with an ST prefix and a 1-2 digit suffix.
Value Type	Number
Format	1-4 digits & 1 letter (if necessary); OG & 2-5 letter prefix, 1-4 digits, and 1-2 digit suffix
Range	1-1656; OG & 2-5 letter prefix (if applicable)

Year

Unique ID	T10B_IC00002
Attribute Definition	The calendar year representing the annual field program and funding the data are associated with.
Value Type	Date
Format	YYYY

Field Date

Unique ID	T10B_IC00015
Attribute Definition	Day, month, and year data were collected.
Value Type	Date
Format	DD-Mon-YY
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Field Crew Member(s)

Unique ID	T10B_IC00379
Attribute Definition	Initials of the field technician(s) who collected data.
Value Type	Text
Format	2 or 3 letters (UPPERCASE) and 1 number (if necessary); 1 set of initials or a combination of many
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Quadrant

Unique ID	T10B_IC00262
Attribute Definition	Quadrant where core or cookie was collected from tree species to measure age.
Value Type	Code
Code	1 ha = 1 hectare plot NE = Northeast Quadrant SE = Southeast Quadrant SW = Southwest Quadrant NW = Northwest Quadrant N = Northern Quadrants S = Southern Quadrants

Tree Core Type

Unique ID	T10B_IC00349
Attribute Definition	Tree cores taken for: Largest Tree i.e. tree with largest DBH found within the central 1 ha of the ABMI site; Leading Species i.e. dominant or co-dominant trees that make up the majority of the stand composition (most abundant species), not including veterans found within each of the four quadrants, and; Secondary Species i.e. dominant or co-dominant trees that make up the second most abundant species but must be $\geq$ 20% of the stand composition, not including veterans found in each of the four quadrants.
Value Type	Code
Code	Largest Tree Leading Species Secondary Species

Common Name

Unique ID	T10B_IC00122
Attribute Definition	Common name of tree from which a core or cookie was taken.
Value Type	Text

Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable NONE = Absent from Survey Area PNA = Protocol Not Available SNI = Species Not Identified SNR = Species Needs Review UID = Unable to Identify
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Scientific Name

Unique ID	T10B_IC00286
Attribute Definition	Scientific name of tree from which a core or cookie was taken.
Value Type	Text
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable NONE = Absent from Survey Area PNA = Protocol Not Available SNI = Species Not Identified SNR = Species Needs Review UID = Unable to Identify

Taxonomic Resolution

Unique ID	T10B_IC00310
Attribute Definition	Resolution to which taxa was identified (e.g. Family, Genus, Species etc.).
Value Type	Text
Missing Values	VNA = Variable Not Applicable

Unique Taxonomic Identification Number

Unique ID	T10B_IC00357
Attribute Definition	Globally unique identifier of tree from which a core or cookie was taken. Unique taxonomic identifiers are generally taken from the International Taxonomic Information System (ITIS; http://www.itis.gov/).
Value Type	Number
Missing Values	VNA = Variable Not Applicable

Significant Tree Damage

Unique ID	T10B_IC00294
Attribute Definition	Any significant damage that could affect the growth of the tree from which the core was taken.
Value Type	Code
Code	BT = Broken Top DT = Dead Top FC = Fork/Crook M = Mistletoe S = Scarring O = Other (indicate damage in comments) None = No Significant Tree Damage

Missing Values DNC = Did Not Collect | PNA = Protocol Not Available | VNA = Variable Not Applicable

Terrestrial T10C Tree Core Rings (since 2016) Raw Data Metadata

Rotation

Unique ID	T10C_IC00001
Attribute Definition	Reference describing when data was collected at a broad level (i.e., Prototype, Rotation 1, Rotation 2). The rotation number indicates the visit number (i.e. 1 = first visit, 2 = second visit or first revisit).
Value Type	Text
Code	Prototype = 2003--2006 Rotation 1 = 2007 onward Rotation 2 = 2015 onward

ABMI Site

Unique ID	T10C_IC00003
Attribute Definition	Reference number given to each ABMI data collection site. An alphabetized suffix distinguishes a new site from the old site(s). Off grid data collection sites are appended with an OG prefix, 2-5 letter project code prefix, and a 1-2 digit suffix. Short term data collection sites are appended with an ST prefix and a 1-2 digit suffix.
Value Type	Number
Format	1-4 digits & 1 letter (if necessary); OG & 2-5 letter prefix, 1-4 digits, and 1-2 digit suffix
Range	1-1656; OG & 2-5 letter prefix (if applicable)

Year

Unique ID	T10C_IC00002
Attribute Definition	The calendar year representing the annual field program and funding the data are associated with.
Value Type	Date
Format	YYYY

Field Date

Unique ID	T10C_IC00015
Attribute Definition	Day, month, and year data were collected.
Value Type	Date
Format	DD-Mon-YY
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Field Crew Member(s)

Unique ID	T10C_IC00379
Attribute Definition	Initials of the field technician(s) who collected data.
Value Type	Text
Format	2 or 3 letters (UPPERCASE) and 1 number (if necessary); 1 set of initials or a combination of many
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Identification Date

Unique ID	T10C_IC01071
Attribute Definition	Day, month, and year data was analyzed by specialist.
Value Type	Date
Format	DD-Mon-YYYY
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Identification Analyst

Unique ID	T10C_IC01054
Attribute Definition	Initials for the technicians/specialists identifying the specimens.
Value Type	Code
Format	2-3 letter code (UPPERCASE) and 1 digit if necessary
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Quadrant

Unique ID	T10C_IC00262
Attribute Definition	Quadrant where core or cookie was collected from tree species to measure age.
Value Type	Code
Code	1 ha = 1 hectare plot NE = Northeast Quadrant SE = Southeast Quadrant SW = Southwest Quadrant NW = Northwest Quadrant N = Northern Quadrants S = Southern Quadrants

Tree Core Type

Unique ID	T10C_IC00349
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Attribute Definition	Tree cores taken for: Largest Tree i.e. tree with largest DBH found within the central 1 ha of the ABMI site; Leading Species i.e. dominant or co-dominant trees that make up the majority of the stand composition (most abundant species), not including veterans found within each of the four quadrants, and; Secondary Species i.e. dominant or co-dominant trees that make up the second most abundant species but must be $\geq 20\%$ of the stand composition, not including veterans found in each of the four quadrants.
Value Type	Code
Code	Largest Tree Leading Species Secondary Species

Sample Type

Unique ID	T10C_IC00285
Attribute Definition	Indicates whether a core or cookie type of sample was taken
Value Type	Code
Code	Cookie Core
Missing Values	DNC = Did Not Collect PNA = Protocol Not Available VNA = Variable Not Applicable

Common Name

Unique ID	T10C_IC00122
Attribute Definition	Common name associated with the organism.
Value Type	Text
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable NONE = Absent from Survey Area PNA = Protocol Not Available SNI = Species Not Identified SNR = Species Needs Review UID = Unable to Identify

Scientific Name

Unique ID	T10C_IC00286
Attribute Definition	Scientific name associated with the organism.
Value Type	Text
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable NONE = Absent from Survey Area PNA = Protocol Not Available SNI = Species Not Identified SNR = Species Needs Review UID = Unable to Identify

Taxonomic Resolution

Unique ID	T10C_IC00310
Attribute Definition	Taxonomic resolution to which identification was made (e.g. Family, Genus, Species etc.).
Value Type	Text
Missing Values	VNA = Variable Not Applicable

Unique Taxonomic Identification Number

Unique ID	T10C_IC00357
Attribute Definition	Globally unique identifier of all species. Unique taxonomic identifiers are generally taken from the International Taxonomic Information System (ITIS; http://www.itis.gov/).
Value Type	Number
Missing Values	VNA = Variable Not Applicable

Tree Height (metres)

Unique ID	T10C_IC00350
Attribute Definition	Height of the tree in metres, to the top leaf or stem (0.1 m increments).
Value Type	Number
Format	1 or more digits
Range	1.3 and up
Unit	metre
Missing Values	DNC = Did Not Collect PNA = Protocol Not Available VNA = Variable Not Applicable

DBH (centimetres)

Unique ID	T10C_IC00144
Attribute Definition	Diameter of tree at breast height (1.3 metre) for the largest tree, leading species, or secondary species from which a tree core or cookie was collected to measure age
Value Type	Number
Unit	centimetre
Missing Values	DNC = Did Not Collect PNA = Protocol Not Available VNA = Variable Not Applicable

Number of Rings

Unique ID	T10C_IC01296
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Attribute Definition	A count of the visually distinct tree rings present in the sample.
Value Type	Number
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Number of Estimated Rings

Unique ID	T10C_IC01297
Attribute Definition	An estimate of how many rings are missing before the pith on samples missing the pith and having appropriate rings to make the estimate.
Value Type	Number
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Comment

Unique ID	T10C_IC01300
Attribute Definition	In analyzing tree cores, additional information is provided on the condition and quality of the tree core collected.
Value Type	Code
Code	Bark Missing Broken Complete = Analysis completed Not Analysed = Too many pieces or very unclear rings Pith Missing No Indication of Pith Unclear Pith Unclear Rings Sample Not Mounted
Missing Values	VNA = Variable Not Applicable DNC = Did Not Collect Not Received = Sample not received

Terrestrial T11 Canopy Closure Raw Data Metadata

Rotation

Unique ID	T11_IC00001
Attribute Definition	Reference describing when data was collected at a broad level (i.e., Prototype, Rotation 1, Rotation 2). The rotation number indicates the visit number (i.e. 1 = first visit, 2 = second visit or first revisit).
Value Type	Text
Code	Prototype = 2003--2006 Rotation 1 = 2007 onward Rotation 2 = 2015 onward

ABMI Site

Unique ID	T11_IC00003
Attribute Definition	Reference number given to each ABMI data collection site. An alphabetized suffix distinguishes a new site from the old site(s). Off grid data collection sites are appended with an OG prefix, 2-5 letter project code prefix, and a 1-2 digit suffix. Short term data collection sites are appended with an ST prefix and a 1-2 digit suffix.
Value Type	Number
Format	1-4 digits & 1 letter (if necessary); OG & 2-5 letter prefix, 1-4 digits, and 1-2 digit suffix
Range	1-1656; OG & 2-5 letter prefix (if applicable)

Year

Unique ID	T11_IC00002
Attribute Definition	The calendar year representing the annual field program and funding the data are associated with.
Value Type	Date
Format	YYYY

Field Date

Unique ID	T11_IC01359
Attribute Definition	Day, month, and year data were collected.
Value Type	Date
Format	DD-Mon-YY
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Field Crew Member(s)

Unique ID	T11_IC00379
Attribute Definition	Initials of the field technician(s) who collected data.
Value Type	Text
Format	2 or 3 letters (UPPERCASE) and 1 number (if necessary); 1 set of initials or a combination of many
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Sub-ordinal transect

Unique ID	T11_IC00306
Attribute Definition	Indicates the sub-ordinal transect (NE, SE, SW, NW) along which canopy closure is measured.
Value Type	Code
Code	NE=Northeast Transect SE=Southeast Transect SW=Southwest Transect NW=Northwest Transect

Canopy Cover Location

Unique ID	T11_IC00116
Attribute Definition	First (1) or Second (2) location along the sub-ordinal transects where canopy cover estimates are measured: 1 = 35.35 m stake facing the site centre; 2 = 49.49 m facing away from the site centre. Canopy closure measured at 30 m and 60 m along same transects from 2003-2005 inclusive.
Value Type	Code
Code	1 = 35.35 m stake facing the site centre [30 m from 2003-2005 inclusive] 2 = 49.49 m facing away from the site centre. Canopy closure measured at 30 m and 60 m [60 m from 2003-2005 inclusive]

Canopy Closure (Open dots)

Unique ID	T11_IC00115
Attribute Definition	Canopy closure estimate as a measure of open dots. Canopy closure values range from 0 (high canopy closure) to 96 (low canopy closure).
Value Type	Number
Range	0-96
Missing Values	DNC = Did Not Collect VNA = Variable Not Applicable

Terrestrial Metadata Appendix:

Detailed Code Definitions

Ecosite – Structural Stage

Unique ID	IC00236 in Site Capability Raw Data (RAW_T_SiteCapability)
Unique ID	IC00658 in Shrub Cover--Ecological Site Classification Raw Data (RAW_T_2DCoverSiteChar)
Attribute Definition	Describes the structure stage of the ecosite, as determined after the ecological-site type has been designated i.e. the nutrient/moisture code designation and tree species modifier. The structure stage code starts with a number that defines 1) tree-dominated, 2) non-tree dominated, or 3) open water ecosites. The subsequent letter codes further describe the ecosite structure, in terms of type of vegetation (or dominant substrate type for non-vegetated sites), as well as height and density of vegetation.
Code Definition	1) TS = Tree height is Short TT = Tree height is Tall: D = Dense S = Sparse: C = Complex arrangement N = Non-complex arrangement. 2) N = Non-vegetated: R = substrate of Rock S = substrate of Sand M = substrate of Mineral soil O = substrate of Organic soil; G = Ground vegetation only: B = bryoid/lichen F = forbs G = graminoid R = Reeds and rushes (marsh): D = Dense M = Moderate density S = sparse; S = Shrubs: L = Low height T = Tall; D = Dense M = moderately Dense S = sparse; TL = Low trees; D = Dense M = moderately Dense S = sparse. 3) ON = non-vegetated OV = vegetated: S = short M = medium T = Tall: D = Dense M = Moderate Sparse.

Detailed Code Definition

- I. TSDC = For tree dominated ecosites (trees ≥10% cover), tree height (TS) is Short (≥50% of canopy cover <10 m tall), tree density (D) is Dense (trees ≥1.3 m tall are ≤2 m apart), tree arrangement (C) is Complex (tallest trees ≥10 m apart, with smaller trees - approximately half height - between that receive direct sunlight from above).
- I. TSDN = For tree dominated ecosites (trees ≥10% cover), tree height (TS) is Short (≥50% of canopy cover <10 m tall), tree density (D) is Dense (trees ≥1.3 m tall are ≤2 m apart), tree arrangement (N) is Non-complex (tallest trees <10 m apart, with few or no smaller trees - approximately half height - between that receive direct sunlight from above).

1. TSSC = For tree dominated ecosites (trees $\geq 10\%$ cover), tree height (TS) is Short ($\geq 50\%$ of canopy cover < 10 m tall), tree density (S) is Sparse (trees ≥ 1.3 m tall are > 2 m apart), tree arrangement (C) is Complex (tallest trees ≥ 10 m apart, with smaller trees - approximately half height - between that receive direct sunlight from above).
1. TSSN = For tree dominated ecosites (trees $\geq 10\%$ cover), tree height (TS) is Short ($\geq 50\%$ of canopy cover < 10 m tall), tree density (S) is Sparse (trees ≥ 1.3 m tall are > 2 m apart), tree arrangement (N) is Non-complex (tallest trees < 10 m apart, with few or no smaller trees - approximately half height - between that receive direct sunlight from above).
1. TTDC = For tree dominated ecosites (trees $\geq 10\%$ cover), tree height (TT) is Tall ($> 50\%$ of canopy cover ≥ 10 m tall), tree density (D) is Dense (trees ≥ 1.3 m tall are ≤ 2 m apart), tree arrangement (C) is Complex (tallest trees ≥ 10 m apart, with smaller trees - approximately half height - between that receive direct sunlight from above).
1. TTDN = For tree dominated ecosites (trees $\geq 10\%$ cover), tree height (TT) is Tall ($> 50\%$ of canopy cover ≥ 10 m tall), tree density (D) is Dense (trees ≥ 1.3 m tall are ≤ 2 m apart), tree arrangement (N) is Non-complex (tallest trees < 10 m apart, with few or no smaller trees - approximately half height - between that receive direct sunlight from above).
1. TTSC = For tree dominated ecosites (trees $\geq 10\%$ cover), tree height (TT) is Tall ($> 50\%$ of canopy cover ≥ 10 m tall), tree density (S) is Sparse (trees ≥ 1.3 m tall are > 2 m apart), tree arrangement (C) is Complex (tallest trees ≥ 10 m apart, with smaller trees - approximately half height - between that receive direct sunlight from above).
1. TTSN = For tree dominated ecosites (trees $\geq 10\%$ cover), tree height (TT) is Tall ($> 50\%$ of canopy cover ≥ 10 m tall), tree density (S) is Sparse (trees ≥ 1.3 m tall are > 2 m apart), tree arrangement (N) is Non-complex (tallest trees < 10 m apart, with few or no smaller trees - approximately half height - between that receive direct sunlight from above).
2. NR = For non-tree dominated (trees $< 10\%$ cover), non-vegetated ($< 10\%$ vegetation cover) ecosites, dominant substrate type (NR) is Bedrock, cliff, talus, or boulder
2. NS = For non-tree dominated (trees $< 10\%$ cover), non-vegetated ($< 10\%$ vegetation cover) ecosites, dominant substrate type (NS) is Sand bar in river/stream (cobble, gravel, sand)
2. NM = For non-tree dominated (trees $< 10\%$ cover), non-vegetated ($< 10\%$ vegetation cover) ecosites, dominant substrate type (NM) is Mineral soil due to any other reason
2. NO = For non-tree dominated (trees $< 10\%$ cover), non-vegetated ($< 10\%$ vegetation cover) ecosites, dominant substrate type (NO) is Organic soil
2. GBD = For non-tree dominated (trees $< 10\%$ cover) ecosites, where only ground vegetation (shrubs $< 10\%$; trees $< 10\%$; other vascular plants $> 10\%$) is present, vegetation type (GB) is Bryoid/Lichen (Bryophyte and lichen), vegetation density (D) is Dense (cover $> 75\%$)
2. GBM = For non-tree dominated (trees $< 10\%$ cover) ecosites, where only ground vegetation (shrubs $< 10\%$; trees $< 10\%$; other vascular plants $> 10\%$) is present,

- vegetation type (GB) is Bryoid/Lichen (Bryophyte and lichen), vegetation density (M) is Moderate (cover is between 25-75%)
2. GBS = For non-tree dominated (trees <10% cover) ecosites, where only ground vegetation (shrubs <10%; trees <10%; other vascular plants >10%) is present, vegetation type (GB) is Bryoid/Lichen (Bryophyte and lichen), vegetation density (S) is Sparse (cover is <25%)
 2. GFD = For non-tree dominated (trees <10% cover) ecosites, where only ground vegetation (shrubs <10%; trees <10%; other vascular plants >10%) is present, vegetation type (GF) is Forb (non-graminoid herbs and ferns), vegetation density (D) is Dense (cover >75%)
 2. GFM = For non-tree dominated (trees <10% cover) ecosites, where only ground vegetation (shrubs <10%; trees <10%; other vascular plants >10%) is present, vegetation type (GF) is Forb (non-graminoid herbs and ferns), vegetation density (M) is Moderate (cover is between 25-75%)
 2. GFS = For non-tree dominated (trees <10% cover) ecosites, where only ground vegetation (shrubs <10%; trees <10%; other vascular plants >10%) is present, vegetation type (GF) is Forb (non-graminoid herbs and ferns), vegetation density (S) is Sparse (cover is <25%)
 2. GGD = For non-tree dominated (trees <10% cover) ecosites, where only ground vegetation (shrubs <10%; trees <10%; other vascular plants >10%) is present, vegetation type (GG) is Graminoid (grasses, sedges), vegetation density (D) is Dense (cover >75%)
 2. GGM = For non-tree dominated (trees <10% cover) ecosites, where only ground vegetation (shrubs <10%; trees <10%; other vascular plants >10%) is present, vegetation type (GG) is Graminoid (grasses, sedges), vegetation density (M) is Moderate (cover is between 25-75%)
 2. GGS = For non-tree dominated (trees <10% cover) ecosites, where only ground vegetation (shrubs <10%; trees <10%; other vascular plants >10%) is present, vegetation type (GG) is Graminoid (grasses, sedges), vegetation density (S) is Sparse (cover is <25%)
 2. GRD = For non-tree dominated (trees <10% cover) ecosites, where only ground vegetation (shrubs <10%; trees <10%; other vascular plants >10%) is present, vegetation type (GR) is Marsh (reeds and rushes), vegetation density (D) is Dense (cover >75%)
 2. GRM = For non-tree dominated (trees <10% cover) ecosites, where only ground vegetation (shrubs <10%; trees <10%; other vascular plants >10%) is present, vegetation type (GR) is Marsh (reeds and rushes), vegetation density (M) is Moderate (cover is between 25-75%)
 2. GRS = For non-tree dominated (trees <10% cover) ecosites, where only ground vegetation (shrubs <10%; trees <10%; other vascular plants >10%) is present, vegetation type (GR) is Marsh (reeds and rushes), vegetation density (S) is Sparse (cover is <25%)
 2. SLD = For non-tree dominated (trees <10% cover) ecosites, where shrubs (shrubs >10%; trees <10%) are present, shrub height (SL) is Low (shrubby vegetation <2 m tall), shrub density (D) is Dense (shrubs cover >75%)

2. SLM = For non-tree dominated (trees <10% cover) ecosites, where shrubs (shrubs >10%; trees <10%) are present, shrub height (SL) is Low (shrubby vegetation <2 m tall), shrub density (M) is Moderate (shrubs cover is between 25-75%)
2. SLS = For non-tree dominated (trees <10% cover) ecosites, where shrubs (shrubs >10%; trees <10%) are present, shrub height (SL) is Low (shrubby vegetation <2 m tall), shrub density (S) is Sparse (shrubs cover <25%)
2. STD = For non-tree dominated (trees <10% cover) ecosites, where shrubs (shrubs >10%; trees <10%) are present, shrub height (ST) is Tall (shrubby vegetation >2 m tall), shrub density (D) is Dense (shrubs cover >75%)
2. STM = For non-tree dominated (trees <10% cover) ecosites, where shrubs (shrubs >10%; trees <10%) are present, shrub height (ST) is Tall (shrubby vegetation >2 m tall), shrub density (M) is Moderate (shrubs cover is between 25-75%)
2. STS = For non-tree dominated (trees <10% cover) ecosites, where shrubs (shrubs >10%; trees <10%) are present, shrub height (ST) is Tall (shrubby vegetation >2 m tall), shrub density (S) is Sparse (shrubs cover <25%)
2. TLD = For non-tree dominated (trees <10% cover) ecosites, where small trees (small trees >10%; trees <10%) are present and small tree density (D) is Dense (small trees cover >75%)
2. TLM = For non-tree dominated (trees <10% cover) ecosites, where small trees (small trees >10%; trees <10%) are present and small tree density (M) is Moderate (small trees cover is between 25-75%)
2. TLS = For non-tree dominated (trees <10% cover) ecosites, where small trees (small trees >10%; trees <10%) are present and small tree density (S) is Sparse (small trees cover <25%)
3. ON = For open water dominated ecosites (emergent vegetation <10%), vegetation type (ON) is Non-vegetated
3. OVSD = For open water dominated ecosites (emergent vegetation <10%), vegetation type (OV) is Vegetated (floating and submerged plants $\geq$ 10% cover), vegetation height (S) is Short submerged ($\geq$ 50% of vegetation extends 0.0 - <0.3 m above the substrate), vegetation density (D) is Dense (aquatic vegetation covers >75% of the substrate)
3. OVSM = For open water dominated ecosites (emergent vegetation <10%), vegetation type (OV) is Vegetated (floating and submerged plants $\geq$ 10% cover), vegetation height (S) is Short submerged ($\geq$ 50% of vegetation extends 0.0 - <0.3 m above the substrate), vegetation density (M) is Moderate (aquatic vegetation covers 25-75% of the substrate)
3. OVSS = For open water dominated ecosites (emergent vegetation <10%), vegetation type (OV) is Vegetated (floating and submerged plants $\geq$ 10% cover), vegetation height (S) is Short submerged ($\geq$ 50% of vegetation extends 0.0 - <0.3 m above the substrate), vegetation density (S) is Sparse (aquatic vegetation covers <25% of the substrate)
3. OVMD = For open water dominated ecosites (emergent vegetation <10%), vegetation type (OV) is Vegetated (floating and submerged plants $\geq$ 10% cover), vegetation height (M) is Medium submerged ($\geq$ 50% of vegetation extends 0.3 - 1.3 m above the substrate), vegetation density (D) is Dense (aquatic vegetation covers >75% of the substrate)

3. OVMM = For open water dominated ecosites (emergent vegetation <10%), vegetation type (OV) is Vegetated (floating and submerged plants $\geq$ 10% cover), vegetation height (M) is Medium submerged ($\geq$ 50% of vegetation extends 0.3 - 1.3 m above the substrate), vegetation density (M) is Moderate (aquatic vegetation covers 25-75% of the substrate) above the substrate)
3. OVMS = For open water dominated ecosites (emergent vegetation <10%), vegetation type (OV) is Vegetated (floating and submerged plants $\geq$ 10% cover), vegetation height (M) is Medium submerged ($\geq$ 50% of vegetation extends 0.3 - 1.3 m above the substrate), vegetation density (S) is Sparse (aquatic vegetation covers <25% of the substrate)
3. OVTD = For open water dominated ecosites (emergent vegetation <10%), vegetation type (OV) is Vegetated (floating and submerged plants $\geq$ 10% cover), vegetation height (T) is Tall submerged ($\geq$ 50% of vegetation extends >1.3 m above the substrate), vegetation density (D) is Dense (aquatic vegetation covers >75% of the substrate)
3. OVTM = For open water dominated ecosites (emergent vegetation <10%), vegetation type (OV) is Vegetated (floating and submerged plants $\geq$ 10% cover), vegetation height (T) is Tall submerged ($\geq$ 50% of vegetation extends >1.3 m above the substrate), vegetation density (M) is Moderate (aquatic vegetation covers 25-75% of the substrate)
3. OVTS = For open water dominated ecosites (emergent vegetation <10%), vegetation type (OV) is Vegetated (floating and submerged plants $\geq$ 10% cover), vegetation height (T) is Tall submerged ($\geq$ 50% of vegetation extends >1.3 m above the substrate), vegetation density (S) is Sparse (aquatic vegetation covers <25% of the substrate)
3. OVFD = For open water dominated ecosites (emergent vegetation <10%), vegetation type (OV) is Vegetated (floating and submerged plants $\geq$ 10% cover), vegetation height (F) is Floating ($\geq$ 50% of vegetation with floating leaves on water surface), vegetation density (D) is Dense (aquatic vegetation covers >75% of the substrate)
3. OVFM = For open water dominated ecosites (emergent vegetation <10%), vegetation type (OV) is Vegetated (floating and submerged plants $\geq$ 10% cover), vegetation height (F) is Floating ($\geq$ 50% of vegetation with floating leaves on water surface), vegetation density (M) is Moderate (aquatic vegetation covers 25-75% of the substrate)
3. OVFS = For open water dominated ecosites (emergent vegetation <10%), vegetation type (OV) is Vegetated (floating and submerged plants $\geq$ 10% cover), vegetation height (F) is Floating ($\geq$ 50% of vegetation with floating leaves on water surface), vegetation density (S) is Sparse (aquatic vegetation covers <25% of the substrate)